

Control of a superconducting transmon qutrit

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Quantum information processing with d -level systems, or qudits, has garnered significant interest due to their potential advantages over traditional qubit-based computation. Current quantum computing platforms—such as superconducting circuits, trapped ions, Rydberg atoms, and photonic circuits—facilitate access to higher energy states naturally, without additional hardware complexity. This multilevel architecture expands the Hilbert space, offering potential improvements in fault-tolerance, high-fidelity magic state distillation, efficient algorithms, analog simulation, and quantum communication. In this thesis, we explore how to precisely control a qutrit system, a minimal example of qudits ($d = 3$). The qutrit experimentally realized by confining a superconducting transmon circuit to its lowest three eigenstates. We develop the theory of universal control over the Hilbert space of single qutrit realized in a circuit quantum electrodynamics system. Next, we characterize the coherence of a transmon qutrit made cloud-accessible on IBM Quantum Experience. By refining use of the amplified phase error sequence (APE), we find that the dominant error type in our system is phase errors. Finally, combining the APE protocol with Derivative Removal of Adiabaticity (DRAG) correction, we obtain an elementary $\pi/2$ rotation with an error rate reduction of 60%, compared to non-corrected gates. This reduction is then further quantified by performing Clifford-based qutrit randomized benchmarking. The experiment results in an averaged fidelity $F = 98.9\%$ for the single-qutrit Clifford gates, limiting by the incoherent nature of the charge-sensitive transmon system.

Control of a superconducting transmon qutrit

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This thesis is dedicated to my mom.

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Contents

1	Introduction	1
1.1	Motivation	1
1.2	Thesis overview	2
2	Fundamentals of quantum information	3
2.1	ABC of quantum information	3
2.1.1	The atom of computation	3
2.1.2	Single-qubit gates	5
2.1.3	Measurement	6
2.1.4	Entanglement	7
2.2	Quantum error correction	8
2.2.1	Repetition code: classical vs. quantum	8
2.2.2	Stabilizer formalism	12
2.2.3	Toric code	14
2.2.4	The Knill-Laflamme condition	20
2.3	Moving higher: qutrit systems	20
2.3.1	Properties of a qutrit	21
2.3.2	Single-qutrit gates: theory	22
2.3.3	A toy example: speedup in parity check using qutrit	25
3	Circuit quantum electrodynamics	27
3.1	Introduction	27
3.1.1	The quantum harmonic oscillator	27
3.2	Adding nonlinearities: transmon qubit	29
3.2.1	Charge dispersion	30
3.2.2	Anharmonicity of the transmon	31
3.2.3	Coupling to cavity	33
3.2.4	Encoding a qutrit in transmon	34
3.3	Driving a transmon	34
3.3.1	Introducing the drive	34
3.3.2	Single-qutrit gates: realization	35

3.4	Readout transmon	38
3.4.1	Dispersive regime	38
3.4.2	Dispersive readout: the non-demolition aspect	39
3.4.3	Qutrit readout on transmon	40
4	Characterization of a transmon qutrit	42
4.1	Qutrit spectroscopy	42
4.2	Rabi oscillation	44
4.3	Ramsey experiment	46
5	Incoherent-limited control and benchmarking	49
5.1	Errors in a superconducting qutrit	50
5.1.1	Phase errors	50
5.1.2	Measuring phase errors	51
5.1.3	Correcting phase errors using DRAG	51
5.1.4	Over/under-rotation errors	52
5.2	Randomized benchmarking	54
5.2.1	Qubit randomized benchmarking	54
5.2.2	Qutrit randomized benchmarking	56
6	Conclusion	58
	Bibliography	58
	Appendices	63
A	Proof of Mathematical Theorems	64
A.1	Parity of a permutation	64

1.1 Motivation

Quantum information processing using d -level systems has attracted increasing attention recently due to demonstrated advantages over the usual approach that utilized two-level systems or qubits as basic unit of computation. The multilevel structure of current quantum computing platforms, including superconducting circuits [21, 5, 13, 4, 9], trapped ions [32], Rydberg atoms [18], photonic circuits [25, 20], allows natural access to higher energy states without incurring hardware overhead. This results in a larger, interconnected Hilbert space with potential benefits in improved fault-tolerance [7] and high-fidelity magic state distillation [8], efficient algorithms [16, 17], analog simulation [29, 6], and communication [42].

Three-level systems, i.e. qutrits are the simplest case in which quantum speedup could be found over their qubit counterparts [16, 34]. In superconducting quantum devices, such system can be realised by confining the transmon circuits to their lowest three energy levels. The system's anharmonicity, typically on order of a few hundreds MHz, is sufficient for selective coherent control of the subspaces defined by the eigenstates $\{|0\rangle, |1\rangle\}$ and $\{|1\rangle, |2\rangle\}$ by microwave pulses. Nonetheless, the multilevel nature of transmons that allows exploration of beyond-qubit computational advantage comes at the expense of unwanted interference with higher states (e.g. the $|3\rangle$ state) that leads to manifold leakage and enhanced dephasing [31]. Previous experiments have shown that by switching to a higher E_J/E_C ratio design ≥ 50 , one can mitigate such effects on transmon systems and thereby suppressing qutrit control infidelity to below 10^{-3} , measured by interleaved randomized benchmarking (IRB) [27]. The design change, however also comes at a cost since the transmon's anharmonicity reduces with power law as E_J/E_C increases. Alternative proposals have also been made to explore the multilevel approach, including optimal control methods [38] and intrinsic noise-protected superconducting circuits [46, 22]; however these schemes often require complex experimental design and thus complicates the accompanied driving schemes. Concomitant with the predicted breakdown of ideal current-phase relations [45] and quasiparticle poisoning [33], they present fundamental limitations in Josephson-junction-based quantum information processing devices using circuit QED

architecture.

The gate fidelity is an important figure of merit to benchmark how far away we are from the fault-tolerant quantum computing era. Depending on the error correcting codes and the underlying circuit noise model, the gate fidelity threshold varies between 10^{-2} to 10^{-4} [40, 14, 41] (qubit case). For qutrit-based fault-tolerant quantum computers, determining this number is an open question. In this thesis, we try to push the single-qutrit gate fidelity F as high as possible, so that given a qutrit-code, it is likely to be good enough to be used as a set of physical gates. The upper bound for single-qutrit gate fidelity is the incoherent limit. Since each single-qutrit gate, as we shall see, is decomposed into a sequence of subspace rotations, the incoherent limit is ultimately established by the coherence time of the subspaces that form the computational subspace of the qutrit, namely $\{|0\rangle, |1\rangle\}$ and $\{|1\rangle, |2\rangle\}$. In this thesis, we adopt a pulse-level noise analysis that takes into account physical errors presenting in a circuit quantum electrodynamics system. The pulse-level noise model is then quantitatively verified and corrected. Finally, the overall incoherent-limited gate set realized from error-corrected $\pi/2$ pulses and virtual Z rotations is benchmarking using qutrit randomized benchmarking. Our work sheds light on the nature of these pulse-level errors, namely phase errors and over/under-rotation and paves the way for future quantum computing with higher energy states experiments.

1.2 Thesis overview

We start with the basic concepts of quantum information and make a small detour to quantum error correction in Chapter 1; we also lay out the theoretical description of a qutrit and how to manipulate its wavefunction. Chapter 3 introduces the vast field of circuit quantum electrodynamics, which is the backbone of today's first generation of quantum computers made out of superconducting qubits. We outline the basic properties of the transmon circuit. We also present the theory of controlling a transmon, and by connecting to the theory described in the first chapter, we present how universal control is obtained over a superconducting qutrit. In Chapter 4 we delve into the experiments where a transmon qutrit made available by the IBM Quantum Experience program is characterized using tools from AMO physics, namely qutrit spectroscopy, Rabi oscillation and Ramsey interferometry. From these experiments, we estimate that the incoherent limit of our single-qutrit gateset is set by the coherence of the $\{|1\rangle, |2\rangle\}$ subspace. Finally, Chapter 5 presents how this work pushes the fidelity of single-qutrit gates up to $F = 98.9\%$ by judicious use of phase error amplification (APE) sequence and Derivative Removal of Adiabaticity (DRAG) correction. The fidelity improvement is then confirmed by qutrit randomized benchmarking.

Fundamentals of quantum information

Quantum information processing generally involves systems made up of one or more quantum bits, or qubits. A qubit is far more complex than a classical bit. While a classical bit represents a single piece of information, a qubit is characterized by two real numbers, effectively holding an infinite amount of information. Moreover, the information required to describe a system of multiple qubits increases exponentially with the number of qubits. The first section will cover the fundamental tools used to describe and manipulate this quantum information. We next spend some time to explore one of the most magical theory in quantum information: quantum error correction (QEC). It is widely believed that without QEC, we cannot build a fault-tolerant quantum computer. Lastly, we return to a general theme of quantum information where the formalism describe a qutrit (i.e. three-level system) in quann mechanics. We also present one of the most important results of this thesis, the universal control over a qutrit.

2.1 ABC of quantum information

2.1.1 The atom of computation

A quantum bit, or in short, a *qubit* is the fundamental unit of quantum information. Any quantum mechanical system with (more than) two states can be a qubit. The states, abstractly speaking, are denoted as $|0\rangle$ and $|1\rangle$. Information is then encoded in the amplitudes of each state,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad (2.1)$$

where α and β are complex and their absolute values sum to unity, $|\alpha|^2 + |\beta|^2 = 1$. The vector $|\psi\rangle$ is formally known as a wavefunction that lives in a two-dimensional Hilbert space (i.e. a complete inner product space). Since α and β are complex-defined, the wavefunction can be rewritten as $|\psi\rangle = |\alpha|e^{i\phi_\alpha}|0\rangle + |\beta|e^{i\phi_\beta}|1\rangle$. Using the Born's rule, it follows that measurement results in the collapse of the wavefunction to a classical state of either $|0\rangle$ or $|1\rangle$ with probability of $|\alpha|^2$ and $|\beta|^2$ respectively. The classical state will

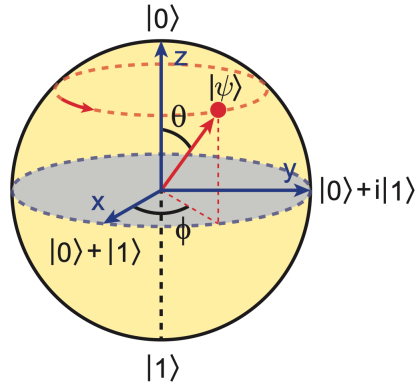


Figure 2.1: The Bloch sphere. A single-qubit (pure) state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is mapped onto a point on the surface of the Bloch sphere. In this case, the state is completely characterized by two real parameters $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle$. A global phase factor $e^{i\phi_{global}} = e^{i(\phi_\alpha - \phi_\beta)} |\psi\rangle$ is irrelevant in this picture.

remain unchanged upon subsequent measurements.

Superposition

Equation 2.1 implies that the qubit may exist in a superposition of states. All superpositions are not the same, although they may do superficially. For example, the states

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad \text{and} \quad \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad (2.2)$$

cannot be resolved in a measurement as the probabilities of measuring $|0\rangle$ and $|1\rangle$ are equal. However, they are different states, fundamentally. In fact they are distinct eigenstates of the $\hat{\sigma}_x$ operator. Together they form a dual basis (in addition to $|0\rangle$ and $|1\rangle$, which are eigenstates of the $\hat{\sigma}_z$ operator) for any wavefunction in a two-dimensional Hilbert space.

To better illustrate the concept of information encoded in a qubit, we make use of a geometrical construction known as the Bloch sphere. A qubit's state is therefore, mapped to a single point on the surface of the sphere. We can reduce the number of real parameters needed to specify the wavefunction by noting that $|\alpha|^2 + |\beta|^2 = 1$ and the global phase, defined by $\phi_{global} = \phi_\alpha - \phi_\beta \equiv$ has no physical meaning. The state is then a vector from the origin to the surface (see Fig. (2.1), uniquely defined by

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle \quad (2.3)$$

where $\theta \in [0, \pi]$ are $\phi \in [0, 2\pi)$ are the azimuthal and equatorial angles of the vector. Note that $\phi = \phi_\beta - \phi_\alpha$. The north/south poles of the sphere correspond to the $|0\rangle$ and $|1\rangle$ states and any other point is a specific superposition of them. The wavefunction is sometimes said to be a two-component vector, $\psi = [\cos(\theta/2), e^{i\phi} \sin(\theta/2)]$.

2.1.2 Single-qubit gates

The next question that one might ask is how to manipulate the state such qubits. The qubit states are nothing but two-component vectors defined on a complex field, $\mathcal{C}$. This means that the *action* on them are represented by 2×2 matrices. In quantum mechanics, these matrices satisfy the Hermitian condition, i.e. $\hat{M} = \hat{M}^\dagger$. A very powerful tool to represent such matrices being the set of Pauli matrices,

$$\hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (2.4)$$

where $\hat{\sigma}_x$ flips the state of the qubit from one to another (quantum analog of a NOT gate), $\hat{\sigma}_z$ turns the relative phase of the qubit from 0 to $e^{i\pi}$ and conversely, and $\hat{\sigma}_y$ does both the jobs. The Pauli matrices are also unitary, i.e. $\hat{\sigma}_j^2 = \hat{I}$ and obey the following commutation relations $[\hat{\sigma}_j, \hat{\sigma}_k] = 2i\epsilon_{jkl}\hat{\sigma}_l$, where the structure constant ϵ_{jkl} is the Levi-Civita symbol and Einstein summation notation is used. In the Bloch sphere picture, Pauli matrices also represent discrete rotations of the Bloch vector around their respective axis. For example, starting from an equal superposition $(|0\rangle + |1\rangle)/\sqrt{2}$, the $\hat{\sigma}_y$ maps this state to $(|0\rangle - |1\rangle)/\sqrt{2}$, which can be further understood as the compound action of $\propto \hat{\sigma}_x\hat{\sigma}_z$ (note that $\hat{\sigma}_x\hat{\sigma}_z = -i\hat{\sigma}_y$). Pictorially, this corresponds to a 180 degrees rotation around the y -axis.

The Pauli operators also form a complete basis for any 2×2 Hermitian matrix $\hat{H} = \frac{1}{2}(H_0\hat{I} + H_x\hat{\sigma}_x + H_y\hat{\sigma}_y + H_z\hat{\sigma}_z)$. In what follows we will see that this formulae comes in handy when describing *continuous* rotations and arbitrarily-defined axis, which is the ingredient that we lack now to reach *any* given point on the Bloch sphere.

Schrodinger equation

In quantum mechanics, the evolution of a quantum mechanical systems can be found according to Schrodinger's equation,

$$i\hbar\frac{\partial}{\partial t}|\psi\rangle = \hat{H}|\psi\rangle, \quad (2.5)$$

where $\hat{H}$ is the Hamiltonian of the system which completely governs the generated dynamics. Such an equation is awfully simple yet complex at the same time. For a time-independent Hamiltonian, the solution is found immediately, $|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi_0\rangle = U|\psi_0\rangle$. The time-independent Hamiltonian can be further decomposed into the sum of Pauli matrices $\hat{H} = \frac{1}{2}\Omega\hat{n} \cdot \hat{\sigma}$, as mentioned above. In the context of gate-based quantum computation, U is a specific gate. More generally, it is a unitary operator, i.e. $\hat{U}\hat{U}^\dagger = \hat{U}^\dagger\hat{U} = \mathbb{1}$. This implies quantum computing is reversible computing (although note that entropy is still increasing somewhere else).

Continuous rotations

Using Schrodinger equation, we see that a finite continuous amount of rotation about any axis can be applied to the qubit's state. Let $\hat{H} = \frac{1}{2}\Omega\hat{n} \cdot \hat{\sigma}$, then turning on this Hamiltonian for t realizing a unitary

$$\hat{U} = e^{-i\hat{H}t} = e^{-i\frac{\Omega t}{2}\hat{n} \cdot \hat{\sigma}}. \quad (2.6)$$

where we have used $\hbar = 1$. This Hamiltonian can be realized, for example, by putting a two-level system (e.g. an atom) in an oscillating magnetic field $B(t)$. The unitary can be Taylor-expanded to

$$U(t) = \sum_{k=0}^{\infty} \frac{1}{k!} \left(\frac{-i\Omega t}{2} \hat{n} \cdot \hat{\sigma} \right)^k, \quad (2.7)$$

$$U(t) = \sum_{k=0}^{\infty} \frac{1}{(2k)!} \left(\frac{-i\Omega t}{2} \right)^{2k} \hat{I} + \sum_{k=0}^{\infty} \frac{1}{(2k+1)!} \left(\frac{-i\Omega t}{2} \right)^{2k+1} (\hat{n} \cdot \hat{\sigma}), \quad (2.8)$$

$$= \cos\left(\frac{\Omega t}{2}\right) \hat{I} - i \sin\left(\frac{\Omega t}{2}\right) (\hat{n} \cdot \hat{\sigma}). \quad (2.9)$$

This implies we can control precisely the amount of rotation driven by the Hamiltonian by either adjusting Ω . Since the Hamiltonian is in unit of energy, Ω is in unit of frequency. In quantum optics, Ω is referred to as Rabi strength (on order of several hundreds MHz) in the above mentioned case, where a two-level system is embedded in an oscillatory field. This results in driving time t on order of several tens of nanoseconds.

Moreover, we can specify the exact rotation axis by tuning the Hamiltonian. As mentioned, any time-independent interaction Hamiltonian can be written as $\hat{H} = \sum_j H_j \hat{\sigma}_j$, where $\hat{\sigma}_j$ is the j -th Pauli matrix. A rotation of angle Ωt around the x -axis, for instance, is realized by setting $H_y = H_z = 0$. Within the superconducting qubit platform, rotations about the x and y axis are straightforward using capacitive X and Y couplings that are readily available. Rotations about the z can be realized by either changing the qubit frequency dynamically (i.e. making $H_z(t)$ time-dependent) or by virtual Z -gate, a crucial concept that will be discussed at length later in Chapter 5.

2.1.3 Measurement

Now that we have done something to the state of the qubit, how do we measure it? Within the scope of this thesis, measurement is projective. That is, a measurement outcome is one of the eigenvalues of an *observable* $\hat{O}$ with probability given by the Born's rule $p(m) = \text{tr}(\hat{O}|\psi\rangle\langle\psi|) = \langle\psi|\hat{O}|\psi\rangle$, where $|\psi\rangle$ is the state of the quantum system. The observable, being Hermitian, has a spectral decomposition,

$$\hat{O} = \sum_m m \hat{O}_m, \quad (2.10)$$

where $\hat{O}_m$ is the projector onto the eigenspace of $\hat{O}$ with its associative eigenvalue m . The state of the system after the measurement is said to be projected onto one of its subspace (could be both computational or non-computational). Given that we readout an outcome m , the post-measurement state of the quantum system is $\hat{O}_m|\psi\rangle/\sqrt{p(m)}$. It is also noteworthy that the set of $\hat{O}_m$ has to be orthogonal projectors, $\hat{O}_m\hat{O}_{m'} = \delta_{m,m'}\hat{O}_m$ and complete, $\sum_m \hat{O}_m^\dagger \hat{O}_m = \hat{I}$.

Although directly measuring the state of the qubit will "collapse" its state into a classical state, we can make clever measurement such that it interrogate a *specific* information of the system, yet leaving the state unperturbed. For example, asking the question "is the qubit in state 0" will inevitable collapse its state to either 0 or 1, while asking "are you in the same state with this qubit j " is allowed. Although there will be back-action associated with such a measurement, this is generally allowed and is referred to as projecting the state onto the so-called "code-space" or "error-space". This type of measurement is one of the most crucial requirement for quantum error correction to work.

Another important aspect of quantum measurement is the "non-demolitioness". A measurement is said to be quantum-nondemolition if during the process of measurement, the uncertainty of the measured observable does not increase as a result of the measurement. In other words, measurement does not change the state of the qubit *immediately* before the measurement. This is important because demolition measurements disturb the prior state, which is *the* state that we wish to characterize. This will also be discussed in length in the section of dispersive readout in circuit quantum electrodynamics in Chapter (3).

2.1.4 Entanglement

Suppose we have two qubits. We make them entangled. This is where everything gets spicy.

"Oops! Spooky!"—*Albert Einstein*

Entanglement, to first order approximation, refers to the fact that the quantum state of one particle can be dependent of the state of another. The most famous example being the Bell state of two entangled particles, $|\phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. When measured $\hat{\sigma}_{A,z}$ and $\hat{\sigma}_{B,z}$ separately, the probability that the first particle is in the state $|0\rangle$ and $|1\rangle$ are equally likely, but if one was to measure one in $|0\rangle$, one would also discover that the other one is also $|0\rangle$. In other words, they are correlated. The Bell state is a non-trivial state belongs to a broader class of entangled states of two particles $|\psi_{AB}\rangle$, where they can not be expressed as the tensor product of the individual states $|\psi_{AB}\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle$. Mathematically, we can define this inseparability using the Schmidt decomposition,

$$|\psi_{AB}\rangle = \sum_j \sqrt{\lambda_j} |U_j\rangle_A |V_j\rangle_B, \quad (2.11)$$

where $\sum_j \lambda_j = 1$ and the set of states $|U\rangle_A |V\rangle_B$ forms a complete orthonormal basis for systems A and B respectively. Any state that can be expressed through the Schmidt decomposition using only non-zero λ_j is said to be separable, or pure. States that cannot

are said to be entangled states.

Entanglement generation

Local operation cannot generate entanglement across the two qubits. What we need instead is a non-local interaction Hamiltonian that does not commute with both of the subsystem Hamiltonian H_A, H_B . Let us look at a specific example, where we have two spin-1/2 systems sitting next to each other having no interaction,

$$\hat{H} = \hat{H}_A + \hat{H}_B = \frac{1}{2}\hat{\sigma}_{z,A} + \frac{1}{2}\hat{\sigma}_{z,B}, \quad (2.12)$$

where it is clear that $[\hat{H}_A, \hat{H}_B] = 0$. Since the two local Hamiltonian commute, it follows that the unitary evolution can be written as $\hat{U} = e^{-i\hat{H}t} = e^{-i\hat{H}_A t} e^{-i\hat{H}_B t} = \hat{U}_A \hat{U}_B$. Let us suppose that we start with a product state $|\psi_{AB}\rangle = |\psi\rangle_A |\psi\rangle_B$, then the state at any t reads $|\psi(t)\rangle = e^{-i\hat{H}t} |\psi_{AB}\rangle = (\hat{U}_A |\psi\rangle_A) \otimes (\hat{U}_B |\psi\rangle_B)$, which has a Schmidt decomposition of non-zero coefficients. The state is therefore pure. Let us now add an energy exchange interaction between the two, $\hat{H}_I = g(\sigma_A^+ \sigma_B^- + \sigma_A^- \sigma_B^+)$. The $\hat{\sigma}^+$ and $\hat{\sigma}^-$ are called the raising/lowering operators, which adds/subtracts one quantization of energy from the system. The total Hamiltonian is now $\hat{H} = \hat{H}_A + \hat{H}_B + \hat{H}_I$. It is now clear that $[\hat{H}, \hat{H}_A] \neq 0$ and $[\hat{H}, \hat{H}_B] \neq 0$. The eigenstates of the new total Hamiltonian is therefore now not simply product states of the eigenstates of the individual Hamiltonians.

The Bell states $|\phi^\pm\rangle$ as mentioned above is a prime example of maximally entangled states. It is generated by the following non-local Hamiltonian, $\hat{H} = g\hat{\sigma}_{x,A} \otimes \hat{\sigma}_{x,B}$, which generates the evolution $\hat{U} = \cos(gt)\hat{I} - i\sin(gt)\hat{\sigma}_{x,A}\hat{\sigma}_{x,B}$. Applying this to the product state $|00\rangle$ for exactly $t = \pi/4g$ will realize a Bell state $|\phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - i|11\rangle)$. The Bell states themselves span the two-qubit Hilbert space. That is, in addition to the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, we could have a *dual* basis spanned by the maximally entangled, mutually orthogonal Bell states, $|\phi^+\rangle, |\phi^-\rangle$ and two other states being $|\psi^\pm\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}$.

2.2 Quantum error correction

The closest thing to magic we have is perhaps quantum error correction: the idea of knowing enough about a quantum state to be able to know whether a certain event had happened without knowing *more* so as not to collapse the state. This section presents the fundamental idea behind this and gives concrete examples, ranging from classical repetition to quantum repetition code, the stabilizer formalism, Toric code and concludes with an overarching criterion for quantum codes—the Knill-Laflamme condition.

2.2.1 Repetition code: classical vs. quantum

We will start classically, then we generalize and make it quantum. Let us consider the notorious classical three-bit repetition code. The idea is simple, we encode bits by repeating

them, i.e. we're spreading the information "*the bit is 0*" into three different bits. This is essentially encoding. The logical 0 is chosen such that three bits sharing the same value 0, and similarly the logical bit 1, $0_L = 000, 1_L = 111$.

Error detection and detection

What could happen to our logical bit? There can be bit flips. They turn 0 to 1 and 1 to 0. In general, $0_L \rightarrow 001/010/100$ or $1_L \rightarrow 011/101/110$

The question is now how do we detect such error(s) and correct for it. One way is to check individual bits to see if they are identical. If they're not, then an error happened. Note that this detection method is only possible if we choose to encode the logical 0 as 000; otherwise, this detection method is not possible. We can choose whatever encoding scheme to our desire, but in some cases it would not be a good encoding scheme, in the sense that we would not have a straightforward measure of detecting error (or presumably correcting error, as we shall see).

Now, how do we correct this error? The answer is to reset all values of the encoded bitstring to the majority value. For example, a bitstring 001 indicates that a bit-flip error occurred, and to correct it we simply reset all bits in the string to the majority value, 0. So $001 \rightarrow 000$, and we have successfully restored the logical 0.

Code distance

It should be noted that there are certain cases where we wouldn't be able to detect nor correct an error. One such case being have more than one flipping event. For instance, $111 \rightarrow 100$. If Alice, who is a researcher working with quantum error correction, looks into the identity of the bitstring, Alice knows an error had happened. But can she correct the error, without knowing that initially it was the logical 1_L ? No, Alice would mistakenly deduces a single bit flip event had happened on the first bit, and eventually erasure the information that "the bit is 1".

The fact that we cannot correct more than 1 bit-flip error presents a fundamental limit on the effectiveness of a code (or a class of codes). This property is characterized by the *code distance*. By definition, code distance is the smallest number of bit flips required to transform any two valid codewords into one another, denoted by d .

In our three-bit repetition code, there has to be three single non-repeated bit flip events to transform 000 to 111 and vice versa. Hence, three-bit repetition code has distance $d = 3$. In general, it means that if Alice looks into the identity, Alice knows "there's errors" up to $d - 1$ flipping events, and Alice could possibly successfully correct $(d - 1)/2$ errors at most.

Classical encoded logical operators

Now that we have some classical information, we would like to do computation with them. Say, apply a logical NOT gate, $\text{NOT}0_L = 1_L \leftrightarrow \text{NOT}(000) = 111$, meaning that

we're applying NOT on individual bits. Note that a set of encoded logical operators is non-trivially dependent on the choice of encoding scheme.

Now we enter the quantum realm, and we bring idea of three-bit repetition code along. We define a logical qubit with the computational subspace spanned by the two logical states $|0\rangle_L = |000\rangle$ and $|1\rangle_L = |111\rangle$, such that the state of the qubit is

$$|\psi\rangle = \alpha|0\rangle_L + \beta|1\rangle_L \quad (2.13)$$

meaning that we have spread the information "*the (logical) qubit is in state $|0\rangle$* " into three physical qubits. The computational subspace $\{|000\rangle, |111\rangle\}$ is called the codespace, reside in which are the error-free encoding of the quantum state $|\psi\rangle$.

In addition to bit-flip errors, in quantum computers we also encounter phase flips. Bit-flip errors are represented by the Pauli X and phase-flip errors are represented by the Pauli Z . Their action on the physical qubits are

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle \quad (2.14)$$

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle \quad (2.15)$$

It is noteworthy that all errors that we care in quantum error correction boils down to the bit flips (now X error for short) and phase flips (now Z errors for short).

Remark 1. *If both X and Z errors can be corrected, then any arbitrary error can be corrected, since together X and Z allows universal control over the Hilbert space of any physical qubit. So, the problem boils down to detecting these two errors, and correcting them.*

Before we get into the cleverness of QEC, let us look at the encoded state in the event of a single bit flip error for completeness. For example, the error is $I \otimes X \otimes I$, then the state reads

$$(I \otimes X \otimes I)|\psi\rangle_L = \alpha|010\rangle + \beta|101\rangle \quad (2.16)$$

The state is now not in the codespace, which means there's an error. But here's where it gets tricky (and exciting and the same time). Alice thought to herself, "so to detect an error we, again, look at the identity of the three physical qubits." Not so easy, because in doing so, we'd eventually collapse the state into one of its basis states. We lose the information. Alice might also consider to copy the state $|\psi\rangle_L$ into multiple copies, but in doing so she would violate the no-clone theorem [See Appendix].

Error detection & correction

As discussed, we cannot measure individual qubits in the computational basis. Instead, we measure a certain symmetry properties of the state. For example, one characteristic properties of the state of a logical qubit $|\psi\rangle_L$ encoded on two physical qubits is *the parity* of the physical pair. Alice cannot ask the question whether the two qubits are both in state

Error\Observable	$Z \otimes Z \otimes I$	$Z \otimes I \otimes Z$	$I \otimes Z \otimes I$
$X \otimes I \otimes I$	-1	-1	+1
$I \otimes X \otimes I$	-1	+1	-1
$I \otimes I \otimes X$	+1	-1	-1

Table 2.1: Error syndrom table. For quantum three-qubit repetition code.

$|0\rangle$. However, she can interrogate "Is your parity odd or even?". This could be done by measuring the observable ZZ . It turns out that quantum mechanics allow this question to be made without projecting the state of the two qubits into their respective computational subspaces. And also state-of-the-art experiments have already demonstrate the ability to measure state parity [11]. Let us explore further the idea of measuring the parity of a pair of qubits. Let us suppose exactly one erroneous flipping event might happened, i.e. the error operators are possibly $X \otimes I \otimes I, I \otimes X \otimes I, I \otimes I \otimes X$. Our job is to detect "whether this had happened" by performing a parity check. This could be done by referencing to the syndrome table, see Table (2.1). What can we deduce from the table? Here's the take-away remark

Remark 2. *If the error anti-commutes with the measuring operator, then the outcome is surely -1 .*

Suppose Alice has 3 qubits, $|q_1 q_2 q_3\rangle$ and she measures $\langle ZZI \rangle = +1$. What does it mean? No error, or something happened with the third qubit, but that's not certain. What if she measured $\langle ZZI \rangle = -1$? Then she knows something's off with the first two qubits. But she wouldn't know which is the erroneous one, b.c either $X \otimes I \otimes I$ or $I \otimes X \otimes I$ might have happened. She would have to measure another ZZ . In fact, any $Z_i Z_j$ other than $Z_1 Z_2$. Say this time she measures the first one and the third one, i.e. $\langle Z_1 Z_3 \rangle = -1$, then she would be able to conclude that the error operator is $I \otimes X \otimes I$ (see the table). It is clear that the set of observables has to be cleverly chosen so that one can perform error check in a resource-efficient way.

Now that Alice knew what kind of error has happened, can she resets all the qubits to the majority value? No, because that will require the knowledge of "what's the majority value", i.e. collapsing the information. However, as opposed to classical error correction, where we know "yes, there's a certain error", in the quantum case we know "what specific error has happened". This is extra, and we can use this extra information: the bit-flip X errors are Pauli errors, meaning that they can be reversed by applying the same Pauli operator. This is because all Paulis are self-inverse $\sigma_j^2 = \hat{I}, \forall j$. Hence, all we need to do is applying the same error operator on the logical qubit to correct the error.

Logical operators

Within a certain frame of encoding came with a set of logical operations on the encoded states. In our case, quantum three-qubit repetition code, we want logical $\bar{X}$ and $\bar{Z}$ such

that

$$\bar{X}|0\rangle_L = |1\rangle_L, \quad \bar{X}|1\rangle_L = |0\rangle_L \quad (2.17)$$

$$\bar{Z}|0\rangle_L = |0\rangle_L, \quad \bar{Z}|1\rangle_L = -|1\rangle_L \quad (2.18)$$

One possible choice of $\bar{X}$ and $\bar{Z}$ could be

$$\bar{X} = XXX, \quad \bar{Z} = ZII \quad (2.19)$$

Notice that we can have an odd number of physical Z gates, but not even. This should be evident from the fact that any even number of physical Z gate will result in a logical phase-flip on the logical level.

Code distance

Now we familiarize ourselves with the notion of the distance of a quantum code. But first, the concept the weight of an operator: it's the number of qubits that the operator acts non-trivially on. So for example, $\bar{X} = XXX$ is weighted 3, while $\bar{Z} = ZII$ is weighted 1. Then, we have the definition of the distance.

Remark 3. *Distance of a (quantum) code: the minimal weight of any (non-identity) encoded logical operator comes with the code.*

So, for three-qubit repetition code, the minimal weight is $\bar{Z}$, hence three-qubit repetition code has a distance of 1.

We have good reasons to define something like the code distance. The concept is closely related to the Hamming distance, but going into that direction is beyond the scope of this thesis. For the sake of clarity, if an error (an operator) has the weight less than the code's distance, then it would take the logical qubit out of codespace and we'd be able to detect it; however, if the error's weight is equal to the distance, then essentially what's happened is that we've moved from one codeword (a valid bitstring) to another, i.e. we wouldn't be able to detect it, even though it happened. By definition, the quantum three-qubit repetition code can detect 2 X error and no Z error. This is because any two X error has weight less than the code distance (and three errors make it a valid logical operator); and Z errors are undetectable within our encoding scheme.

2.2.2 Stabilizer formalism

There's a thing called "stabilizer formalism" allowing one to study quantum error correcting codes in a systematic way, e.g. classification of distance or the kinds of errors the code can shield. Also, it can also be used to derive encoded logical operators, which is very important.

Any code that can be defined in the stabilizer formalism is called a stabilizer code. A stabilizer code is defined by specifying two sets of operators. First being the (*stabilizer*) *generators*; and second *the encoded logical operators*.

First, let us talk about the stabilizer group. Let $\{|\psi_j\rangle\}_j$ the codeword basis states ($j^\otimes$ -dimensional Hilbert space). Then the stabilizer group $\mathcal{S}$ is the set of Pauli operators that leave all codeword basis states $|\psi_j\rangle$ invariant.

$$P_k|\psi_j\rangle = |\psi_j\rangle, \forall P_k \in \mathcal{S} \quad (2.20)$$

One would realize that the set of stabilizer must commute, in order for them to leave the codeword basis state invariant. Here's a quick proof by contradiction. Suppose that two stabilizer operators (they are Pauli) don't commute, then $S_1S_2 = -S_2S_1$, which leads to

$$S_1S_2|\psi_j\rangle = |\psi_j\rangle S_2S_1|\psi_j\rangle = -S_1S_2|\psi_j\rangle = -|\psi_j\rangle. \quad (2.21)$$

However, since $S_2S_1 \in \mathcal{S}$ group, then it must be that $S_2S_1|\psi_j\rangle = |\psi_j\rangle$, which is a contraction. Hence, any two operators within a stabilizer group $\mathcal{S}$ must commute.

In group theory, a group G is defined using a set of generators $\{g_j\}_{j=1}^m$. The set order, denoted by $|G|$, is specified by the number 2^m . There's a useful result that links the number of logical qubits k , physical qubits n , and the number of stabilizer generators needed to generate the stabilizer group, which states that

$$n - m = k \quad (2.22)$$

For three-qubit repetition code, $k = 1, n = 3$, and hence we need $3 - 1 = 2$ generators. Where can we look for them: the Pauli group. Can we choose X and Z as generators? No, because they don't commute at all ($XZ = -ZX$ anyway) so they cannot stabilize our codeword states. It turns out that we can choose I and Z , then multiply them out to $2^2 = 4$ (the order of the group).

$$ZZI, ZIZ, IZZ, III \quad (2.23)$$

We've seen a hint of using these operators to detect errors (specifically bit-flip errors) above. By measuring the group of m (m measurements) stabilizer operators, one can detect errors on stabilizer codes. Since $m = n - k$, the number of measurements one have to make scales linearly with the number of physical qubits used for the encoding scheme, which is moderately great.

Encoded logical operators: revisited

We've known that $\bar{X} = XXX$ and $\bar{Z} = ZII$. Are there other choices of the encoded logical operators? There exists two more choices of $\bar{Z}$, for example

$$\bar{Z} = \{ZII, IZI, IIZ, ZZZ\} \quad (2.24)$$

Using the stabilizer formalism, one can actually characterize this set of $\bar{Z}$. First let us see the reason why we have this equivalency. Let $\mathcal{S}$ be the stabilizer group, $|\psi\rangle$ be a state in the codespace, L be a logical operator. Then essentially all logical operators are different

up to a stabilizer,

$$S_j|\psi\rangle = |\psi\rangle, \quad (2.25)$$

$$LS_j|\psi\rangle = L|\psi\rangle \quad (2.26)$$

so essentially what you can have is $ZIIS_j$ being equivalent to ZII , for any $S_j \in \mathcal{S}$, e.g. $(Z \otimes I \otimes I)(Z \otimes Z \otimes I) = (Z^2 \otimes Z \otimes I)$, and because Pauli are self-inverse, $Z^2 = I$, and we obtain $I \otimes Z \otimes I$. It turns out earlier that in order for an operator to become the encoded logical operator L , it has to commute with the (entire) stabilizer group $\mathcal{S}$,

$$[L, S_j] = 0, \forall S_j \in \mathcal{S} \quad (2.27)$$

This means given a logical operator L , there exists a family of $|\mathcal{S}| = 2^m$ operators $\{LS_j\}_j$ that acts equivalently on the codespace. Note that we're looking at the Pauli group G (for n -qubits). The stabilizer $\mathcal{S}$ is a subgroup of G . In group theory, the set of operators in G that commute with every element $S_j \in \mathcal{S}$ is the centralizer $C_G(H)$ of $\mathcal{S}$.

$$C_G(H) = \{C \in G \mid \forall S_j \in \mathcal{S}, CS_j = S_jC\} \quad (2.28)$$

So what we have to do to find the encoded logical operators is to look at the centralizer of $\mathcal{S}$.

2.2.3 Toric code

Discovered by Kitaev in 1997, toric codes are designed to protect a quantum circuits whose qubits are placed on a lattice, which wrapped on a torus. Each edge of the lattice is a qubit. A torus has a periodic boundary condition. Let us first imagine a circle, for any function f defined on the circle it follows that

$$f(\theta) = f(\theta + 2\pi), \quad (2.29)$$

so the function repeats itself. Now imagine a torus, when cutting it at one cross section, we are left with a two-dimensional surface. This two-dimensional surface is the lattice we're talking about. Because the lattice is wrapped on the torus, it follows that even though we're having a $L \times L$ lattice, there's only $2L^2$ edges since the lower (right) gray edges are being glued to the upper (left) edges. We use s to label the 'star' of four bonds emanating from a lattice site (or vertex), j to label an individual bond and its associated physical qubit, and p to label a square (or plaquette). Because each edge is a qubit, the toric code has therefore $n = 2L^2$ physical qubits. The lattice Hamiltonian is therefore assumed the form,

$$\hat{H} = -\sum_s A_s - \sum_p B_p, \quad (2.30)$$

where the A_s and B_p are

$$A_s = \prod_{j \in S} \hat{\sigma}_j^z, \quad B_p = \prod_{j \in P} \hat{\sigma}_j^x \quad (2.31)$$

As we have seen above, to find the stabilizer group for a code, we need to define a set of generators. For toric code, the two generators are *plaquette generator* and *vertex generator*. Since we have two generators, it follows that the order of the group $\mathcal{S}$ is 4. As the name suggests, the plaquette generator is a tensor product of operators associated with a plaquette. Remember that one qubit represents one edge, hence the four qubits enclosing one plaquette P are the ones that being applied. Let Z_S be the operator applying on one qubit, the plaquette generator is

$$\bigotimes_{S \in P} Z_S \quad (2.32)$$

Similarly, the vertex generator is a tensor product of operators associated with a vertex. The four qubits joining in to make a vertex are the ones being applied. Let X_S be the operator applying on one qubit, the vertex operator is

$$\bigotimes_{S \in V} X_S \quad (2.33)$$

One might try and see if the vertex operators and plaquette operators commute. They do, indeed, since they are stabilizers. If the two vertex & plaquette do not overlap on the lattice, then it's trivial to see that the order of operation is irrelevant. If the two vertex & plaquette do overlap, say on qubits q_i and q_j , then

$$(X_i \otimes X_j)(Z_i \otimes Z_j)|q_i q_j\rangle = (X_i Z_i)|q_i\rangle \otimes (X_j Z_j)|q_j\rangle \quad (2.34)$$

$$= -Z_i X_i|q_i\rangle \otimes -Z_j X_j|q_j\rangle \quad (2.35)$$

$$= (Z_i \otimes Z_j)(X_i \otimes X_j)|q_i q_j\rangle \quad (2.36)$$

this means that $[X_i X_j, Z_i Z_j] = 0$.

Plaquette and vertex operators

Let us now consider two adjacent plaquette operators. Question: what are the qubits being applied if we multiply these adjacent operators together? It's the qubits lying on the joint boundary of the two plaquettes. Those who lie on the shared boundary will not be affected, since $Z^2 = I$. The resulting operator will apply on 6 qubits. More generally, for any two plaquettes, if we multiply them together will result in (1) the cancellation of Z 's at the shared boundary and (2) Z 's around the overall boundary. The plaquettes are not necessarily square plaquettes. You can start by a square one, then joining them repeatedly to generate an arbitrarily shaped plaquettes, and the arguments above still hold. Lastly, consider the product of all plaquettes in the entire lattice. Because we're wrapping the lattice around a torus, there's no boundary at all. No qubit is being applied a Z . Hence,

the resulting operator is a one big identity operator,

$$\prod_P \bigotimes_{S \in P}^{L^2} Z_S = I \quad (2.37)$$

This means that the set of all plaquette operators is not an independent set of operators. Note that an independent set of operators means that any operator in the set cannot be expressed as a product of others, i.e. the only solution to $\prod g_j^{a_j} = I$ is $a_j = 0, \forall j$. This concept is strongly analogous to linearly independent, where any vector cannot be a linear combination of the others if they want to form an orthogonal basis.

A stabilizer group cannot be generated from dependent generators. One solution is that we remove one (whatever it is) single plaquette operator from the generator set. Now, if we were to take the product of all plaquette operators on the lattice again, but leaving one single plaquette operator out of the picture, then the product cannot be identity

$$\prod_P \bigotimes_{S \in P}^{L^2-1} Z_S \neq I \quad (2.38)$$

In conclusion, by leaving one single plaquette operator behind, we have $L^2 - 1$ independent plaquette operators (by definition of independent operators).

Dual lattice

Another important concept of the Toric code being the dual lattice. To create a dual lattice from our primal lattice, we shift the primal lattice by half a cell down to the right. The new dual lattice has the same size and same boundary conditions, only that the plaquette are now vertex, and vertex are now plaquette (in terms of the qubits they act on). Because of this, any set of four qubits are being acted on by simultaneously a plaquette/vertex operator in one lattice and vertex/plaquette operator in the dual lattice. You can thus imagine a 90 degree rotation of any edges as the mapping function between the primal & dual lattice. The two lattices are therefore perpendicular.

Remark 4. *A square lattice under such transformation is still square lattice, and we say it is self-dual. On the other hand, the dual lattice of a hexagonal lattice is a triangular lattice.*

We have several good reasons to explore duality of lattice. For instance, multiplication of operators is best understood in the plaquette picture, since all we doing is just dismissing shared boundaries and promoting joint boundaries.

Using lattice duality, the vertex operators in a primal lattice can be defined as the boundary of the corresponding plaquette in its the dual lattice. By the same arguments, we have $L^2 - 1$ independent elements in the set of generators.

$$\prod_V \bigotimes_{S \in V}^{L^2-1} X_V \neq I \quad (2.39)$$

Logical qubits

We have $n = 2L^2$ physical qubits on the torus, $L^2 - 1$ independent plaquette operators and $L^2 - 1$ independent vertex operators, meaning a total of $k = 2(L^2 - 1)$ generators. It then follows that toric code will have $n - k = 2L^2 - 2L^2 + 2 = 2$ encoded logical qubits. Since we have two logical qubits, to complete the code description we find their corresponding encoded logical operators $\mathcal{L} = \{\bar{Z}_1, \bar{Z}_2, \bar{X}_1, \bar{X}_2\}$. First let us remind ourselves with some of the criteria that any $L \in \mathcal{L}$ must satisfy

1. They must commute with all elements of the stabilizer group $\mathcal{S}$.
2. They must not be an element of $\mathcal{S}$ themselves.
3. They must satisfy the commutation and anti-commutation properties of X and Z .

Let us choose a seemingly arbitrary operator, a Z operator acting on a single qubit (edge). This Z commute with all plaquette operators and $Z^2 = 1$. We need to also take care of vertex operators. All vertex operators except those immediately neighboring the edge must commute with the Z because they act on different qubits, but those two vertex immediately adjacent the qubit don't commute with the single Z . Let us consider a pair of adjacent Z 's. Then the two Z 's commute with the one in between them (because of the two minus signs), but is not enough since the left one does not commute. More generally, if we form a string of Z operators, regardless of its shape, there will be one vertex operator at the ends (either left or right) that the string of Z don't commute with that one. A clever way to get over this problem is that you take the right vertex and glue to the left vertex, to form a closed loop of Z 's! Indeed, then we can define an encoded logical $\bar{Z}$ operator that acts on a sequence of edges on the lattice! Let us call this one the $\bar{Z}_1$ operator for the first qubit. Such a loop for $\bar{Z}_1$ cannot be created by a product of plaquette operators. This is because if we have one loop of plaquettes, then the product of them yield two loops of Z 's. That's not one loop of $\bar{Z}$. You may extend the loop of plaquettes, but eventually they will meet because of the periodic boundary, which make them an identity.

We now proceed to define a logical operator $\bar{X}_1$, who must satisfy

1. $[\bar{X}_1, \bar{Z}_1] \neq 0$
2. $[\bar{X}_1, S] = 0, \forall S \in \mathcal{S}$

To find such operator, consider a loop of X operators on the *dual lattice*, perpendicular to the loop defining $\bar{Z}_1$ (intuitively, this is simply because $[X, Z] \neq 0$). This operator must commute with all stabilizer elements, by the same argument one can have with the logical Z operator: Any four qubits associated with a plaquette that overlap with the X line will have exactly two qubits who will yield two minus signs, making them commuting; and vertex operators commute with the loop of X 's by definition. Although we are having these arguments on the dual lattice, we can always transform back to the primal lattice, making the vertices plaquettes, and vice versa. This way, it's easy to see that $\bar{X}_1$ does

not commute with $\bar{Z}_1$, since they overlap on exactly one qubit, resulting in only one minus sign.

We now turn to the second encoded qubit. Again, we seek a pair of $\bar{X}_2$ and $\bar{Z}_2$ which commute with the stabilizers and anticommute with each other. It's quite easy to guess: if we rotate the loops corresponding to $\bar{Z}_1$ and $\bar{X}_1$ 90 degrees, we obtain two new operators which (1) commute with $\bar{Z}_1$ and $\bar{X}_1$ (because the four operators don't overlap on any physical qubits); (2) commute with the plaquettes and vertices (because of the same arguments made above— Z 's commute with themselves and there are two minus signs for XZ); and (3) anticommute with each other (because they share only one physical qubit, and $XZ = -ZX$).

The set of $\mathcal{L}$ described above is not unique, as explored in the three-qubit repetition code case. One can always multiply a $L \in \mathcal{L}$ with a stabilizer $S \in \mathcal{S}$ to get a new, equivalent encoded logical operator. This follows strictly from the fact that the Toric code is a stabilizer code. Notice that the logical operators and stabilizer generators all consist solely of Z 's and X 's. This means in analyzing the equivalency we're can make use of the fact that Pauli are self-inverse extensively. For example, the logical $\bar{Z}$ can be multiply with a (collection) plaquette to form a new logical operator, which can be thought of as an act of deforming the loop wrapped around the torus. Although the loop is diverted from its original path, the overall properties holds. Similarly, the logical $\bar{X}$ are defined on the dual lattice. What's a vertex operator in the primal lattice, due to duality, is a plaquette operator in the dual lattice. The loop for X 's is then also diverted from the original path, but the overall properties holds.

Toric's code distance

The code distance equals to the minimum weight of a encoded logical operator in the code. It's easy to see that the four loops we defined above acting on L physical qubits, hence the code distance $d = L$. In summary, the toric code is denoted as $[[2L^2, 2, L]]$.

In retrospect, one would like to increase the size of the lattice as much as possible, since the larger the code distance, the more likely an error being detected. But there's a limit which we can stop without adding too much redundancy, as we will see later. This limit is called the error correction threshold.

Error detection and correction

An error operator E is a n -qubit Pauli operators, so we can represent them as $E = \{X^{\otimes n}, Y^{\otimes n}, Z^{\otimes n}\}$. We have also established the fact that to detect whether an error has happened, we measure the stabilizers. If we get $+1$, then no error has occurred, otherwise we get -1 , meaning that an error has occurred. The output of the stabilizer measurements, which is a bitstring, is the error syndrome. Notice that the quantum three-repetition code cannot detect the Z error. This is expected, because it is an idea borrowed from classical error correction, in which the Z error does not exist. Toric code, on the other hand, is intrinsically a quantum code with the capacity to correct both X and Z errors.

Let us first consider a $E_Z = Z$ error on a single qubit. To detect such error, we must find a stabilizer that anticommute with E_Z . Remember that we have two kinds of stabilizer operator, the vertices and plaquettes. Plaquettes are Z 's, and hence they commute with E_Z , hence they cannot detect E_Z . Vertex operators on the other hand who adjoin the edge on both sides anticommute with the error. Hence, to detect a single phase flip event on toric code, we measure the output values of two adjacent vertex stabilizers. This idea can be extended to multiple Z errors. First let us consider two Z errors on adjacent qubits. The vertex operator between the two qubits is now commuting with the error $E_Z \otimes E_Z$. We must therefore move to measuring the vertices that adjoin the two ends of the overall edge. In general, given any string of Z errors on the primal lattice, the only stabilizer generators that will detect the errors are the vertices at the two ends of the string. Notice that if the errors occurred on L qubits, then the two vertex operators basically will be identified as one, therefore the errors and the vertex commute, making the errors undetectable. This is in agreement with the fact that toric code's distance is L .

Detection of X errors are carried in the same manner. Strings of X errors on the dual lattice results in -1 outcomes on the plaquette operators adjoin the string on two ends in the dual lattice. Converting this back to the primal lattice, we're measuring the plaquettes. Since X and Z are the generators of Pauli group, and since two classes of errors independently affect different types of stabilizer measurements, we can consider them as two separate error correcting processes. Any insights that we have on one type of error will hold true in the dual lattice of the other error. So, we can focus on one kind of error correction and later we convert it to the dual lattice to complete the description of error correction on toric code.

To correct Pauli errors, we reapply Pauli operators. This is straightforward. The problem lies in where to apply the Paulis, given a certain syndrome.

Error degeneracy

There's a concept of degeneracy in error syndrome. It means that for a given syndrome, there is a certain degree of degeneracy associated with that syndrome, such that the syndrome would indicate faulty conclusion regarding the exact error processes that occurred. In other words, multiple error processes can exhibit the same syndrome. This follows from the fact that two errors E and E' might be different up to any operator L , i.e. $E = E'L$, where L is any operator that commute with the stabilizer.

Let us consider two strings of error that ending at the same vertex operators V_1, V_2 . These two error strings result in the same syndrome $-1, -1$. Suppose Alice concludes E' happened, even though E happened, and applies E' , the resulting operator on the physical qubits is

$$E'E = E'E'L = L \quad (2.40)$$

which, if L is in the stabilizer (stabilizers commute with themselves) would leave the code words invariant. Hence, in this case ($L \in S$), applying E' would correct E . However, L

can be a encoded logical operator, which also commute with $\mathcal{S}$. In this case, the errors can be $E\bar{Z}_1, E\bar{Z}_2, E\bar{Z}_1\bar{Z}_2$, which all lead to the same error syndromes.

2.2.4 The Knill-Laflamme condition

We have seen that some codes can correct for X errors, but cannot correct for Z errors (e.g. quantum three-qubit repetition code). Some, on the other hand, can be proven to be incapable of but only Z errors. The next natural question to ask is why this is the case. This answer to this question is found using the Knill-Laflamme (K-L) condition. Any code must therefore satisfy the K-L condition to be a *intrinsic* quantum error correcting code.

Theorem 1 (Knill-Laflamme condition). *Let C be quantum code, and let P be the projector onto the codespace of C . Suppose $\mathcal{E}$ is a quantum operation with operation elements $\{E_j\}$. The condition for a recovery action $\mathcal{R}$ that maps an erroneous state $\mathcal{E}(\psi)$ back to the codespace is such that,*

$$PE_i^\dagger E_j P = \alpha_{ij} P, \quad (2.41)$$

where α is a Hermitian matrix of complex numbers.

Note that the codespace is completely defined by the rank-1 projector formed by the logical codewords $|W\rangle$, $P = \sum_{\sigma} |W_{\sigma}\rangle\langle W_{\sigma}|$ where $\sigma = \{0, 1\}$. We can therefore rewrite the K-L condition as

$$\langle W_{\sigma} | E_i^\dagger E_j | W_{\sigma'} \rangle = \alpha_{ij} \delta_{\sigma\sigma'}, \quad (2.42)$$

where $\delta_{\sigma\sigma'}$ is the Kronecker delta. The set of operation elements, if such a recover action exists, is then called the set of correctable errors. The proof for K-L condition is well addressed in literature [30].

An important properties of the Hermitian matrix α is its independence on the logical codewords $|W\rangle$ and the structure of its off-diagonal entries. These two properties underpins both the so-called distinguishability and correctability of a quantum code. First, consider the Kronecker delta $\delta_{\sigma\sigma'} = 0$ where

$$\langle W_{\sigma} | E_i^\dagger E_j | W_{\sigma'} \rangle = 0, \forall i, j \quad (2.43)$$

which implies that the states $E_j | W_{\sigma'} \rangle$ and $E_i | W_{\sigma} \rangle$ are orthogonal. This is required because, by measuring the stabilizers, we can distinguish between two scenarios: either the previous state was $|W_{\sigma'}\rangle$ and an error E_j had happened, or the previous state was $|W_{\sigma}\rangle$ and an error E_i had happened.

2.3 Moving higher: qutrit systems

Much of the fundamentals that we developed until now relies on the fact that we can utilise a two-level quantum object, or a qubit, for quantum information processing purposes.

The qubit's wavefunction or state is a vector in a two-dimensional Hilbert space, $|\psi\rangle = a|0\rangle + b|1\rangle$. We have shown that given a generic Hamiltonian $H(t)$ (could be time-independent), the evolution of the state is governed by the Schrodinger equation. The ability to engineering specific Hamiltonian H_{int} of interest, both local and non-local, allow the realization of single-qubit or multiple-qubit gates. Those elements, together with the ability to make measurements to readout the state post-computation, constitute the fundamental building blocks of a quantum information processing device.

However, it is highly motivated that we take an alternative avenue to explore the computational advantage when processing information using the laws of quantum mechanics. One such avenue is using higher energy states of these quantum objects to process information. Using a d -level system, one can realize a single-*qudit* which possesses not just both the prominent effects of quantum mechanics, namely superposition and entanglement, but also the contextuality feature: any realistic hidden-variable theory able to explain the phenomenology of quantum mechanics is contextual for systems of Hilbert space dimension three and greater. Beside this highly foundation-motivated reasons, qudit systems found application in quantum error correction, quantum algorithms and quantum simulation. Qutrit, $d = 3$, is the simplest case in which we observe computational speedup compared to their two-level counterparts [7, 8, 16, 17, 29, 6, 42]. The rest of this thesis will be devoted to two main purposes. Firstly, we deal with the question how to realize a qutrit system on a transmon circuit using circuit quantum electrodynamics; and second, having such a single-qutrit system realized on the transmon circuit, how do we manipulate and control its state with high precision. While the former is answered in Chapter 3, the latter is investigated in Chapter 4 and Chapter 5.

2.3.1 Properties of a qutrit

First and foremost, in this section we lay the foundation for much of the theoretical and experimental procedures described later on. We will develop a framework for a single qutrit system, most prominently the mathematical description of single-qutrit gates and qutrit measurement. The state of a qutrit is specified by a set of three complex-valued numbers, or coefficients, let them be a, b, c ,

$$|\psi\rangle = a|0\rangle + b|1\rangle + c|2\rangle, \quad (2.44)$$

where the sum of their squares is unity (normalization) $|a|^2 + |b|^2 + |c|^2 = 1$. The density matrix of a qutrit, written in the Gell-Mann matrices λ_a , is

$$\rho = |\psi\rangle\langle\psi| = \frac{1}{3}(I + \lambda_a P_a), \quad (2.45)$$

where I is the 3×3 identity element and P_a is a component of a real vector with 8 entries sitting in the adjoint representation of the $SU(3)$ group [3]. It is straightforward to verify that ρ satisfies all the usual conditions, namely (a) being Hermitian; (b) $\text{tr}(\rho) = 1$; (c) positive semidefinite; and (d) $\rho^2 \leq \rho$. Note that for the last condition, the equality implies

ρ is a pure state $\rho^2 = 1$, which consequently can be used to prove that ρ itself is positive semidefinite.

The Gell-Mann matrices are Hermitian and they satisfy the quation

$$\lambda_a \lambda_b = \frac{2}{3} \delta_{ab} I + (d_{abc} + i f_{abc}) \lambda_c, \quad (2.46)$$

where f_{abc} are the structure constants of the SU(3) algebra. For completeness, I list down here the eight Gell-Mann matrices,

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (2.47)$$

$$\lambda_4 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \lambda_5 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \lambda_8^* = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad (2.48)$$

$$\lambda_6 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad \lambda_7 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & -i \\ 0 & 1 & 0 \\ i & 0 & -2 \end{pmatrix}. \quad (2.49)$$

The GM matrices satisfy the commutation relation $[\lambda_a, \lambda_b] = 2i f_{abc} \lambda_c$. Another important properties of the Gell-Mann matrices is that, since they form a complete basis for 3×3 Hermitian matrices, they are also the generators of rotation for the special unitary SU(3) group. We will now use this fact to construct the complete gate set for a single qutrit. The specific local interaction Hamiltonian that gives rise to such dynamics will be specified in Chapter 5.

2.3.2 Single-qutrit gates: theory

Starting from the Gell-Mann matrices, it is then an elementary result that the eneral group element for the fundamental representation of SU(3) is expressed as a second order polynomial in the hermitian generating matrix $H(\lambda_a)$ [12]. Consider an arbitrary traceless Hermitian H , then any SU (3) group element generated by a traceless 3×3 Hermitian matrix H is

$$e^{i\theta H} = \sum_{k=0,1,2} \left[H^2 + \frac{2}{\sqrt{3}} H \sin \left(\theta + \frac{2\pi k}{3} \right) - \frac{1}{3} I \left(1 + 2 \cos \left(2 \left(\phi + \frac{2\pi k}{3} \right) \right) \right) \right] \times \frac{\exp \left(\frac{2}{\sqrt{3}} i \theta \sin \left(\phi + \frac{2\pi k}{3} \right) \right)}{1 - 2 \cos \left(2 \left(\phi + \frac{2\pi k}{3} \right) \right)}, \quad (2.50)$$

where ϕ is a function of the determinant of H , $\phi = \frac{1}{3} \left(\cos^{-1} \left(\frac{3}{2} \sqrt{3} \det(H) \right) - \frac{\pi}{2} \right)$. Here, if we think about Gell-Mann matrices, they represent a spin-1 system with $\det H = 0$.

Consequently, $\phi = 0$. The representation (Eqn. 2.50) is therefore succinctly written as

$$e^{i\theta H} = I + iH \sin \theta + H^2 (\cos \theta - 1). \quad (2.51)$$

This relation (Eqn. 2.51) holds true if H is one of the first seven Gell-Mann matrices specified above in (Eqn. 2.47). The last element, λ_8 is the anomaly where $\phi \neq 0$ and having two degenerate eigenvalues. However, a compact form of the representation $SU(3)$ of such matrix can be avoided as it corresponds to a concatenated ZZ rotation of the (0-1) and (1-2) subspace of the qutrit, which can be found (or realized) by introducing another element that is not in the Gell-Mann set, but of usefulness for further discussion,

$$\lambda_8^* = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad (2.52)$$

which corresponds to solely the rotation about the z -axis in the (1-2) subspace. It is clear that $\phi_8 = \pi/6$ while $\phi_{8*} = 0$ and therefore the compact form of Eqn. 2.51 can be used.

We are now in the position to obtain representation results for the $SU(3)$ elements using the Gell-Mann matrices. These are the 3×3 unitary matrix of the special unitary group $SU(3)$ that corresponds to rotation within the so-called computational subspace that is formally defined by the rank-1 projector $P = |0\rangle\langle 0| + |1\rangle\langle 1| + |2\rangle\langle 2|$, where $|0\rangle$, $|1\rangle$, $|2\rangle$ are the computational eigenstates of the qutrit. In practice, however, due to experimental constraint, *intrinsic* qutrit gates are often hard to realized. The usual workaround is to use the so-called two-level subspace rotations. This means we are interested in unitary matrices that act non-trivially on the two-level subspaces spanned by $\{|0\rangle, |1\rangle\}$, for short (01); and $\{|1\rangle, |2\rangle\}$, for short (12). Within the (01) subspace, they are unitary rotation about the x, y, z axes of the form

$$\mathcal{R}_X^{01}(\theta) = e^{-i\frac{\theta}{2}\lambda_1} = \begin{pmatrix} \cos \theta/2 & -i \sin \theta/2 & 0 \\ -i \sin \theta/2 & \cos \theta/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (2.53)$$

$$\mathcal{R}_Y^{01}(\theta) = e^{-i\frac{\theta}{2}\lambda_2} = \begin{pmatrix} \cos \theta/2 & \sin \theta/2 & 0 \\ -\sin \theta/2 & \cos \theta/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (2.54)$$

$$\mathcal{R}_Z^{01}(\theta) = e^{-i\frac{\theta}{2}\lambda_3} = \begin{pmatrix} e^{-i\theta/2} & 0 & 0 \\ 0 & e^{i\theta/2} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (2.55)$$

Similarly, within the (12) subspace the set of Gell-Mann generators gives rise to unitary rotations about the x, y, z axes. Note that the last operator $\mathcal{R}_Z^{12}$ is generated by not an element of the Gell-Mann set, but as mentioned this matrix is a traceless 3×3 Hermitian

and thus satisfy the condition required by (Eqn. 2.51),

$$\mathcal{R}_X^{12}(\theta) = e^{-i\frac{\theta}{2}\lambda_4} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta/2 & -i \sin \theta/2 \\ 0 & -i \sin \theta/2 & \cos \theta/2 \end{pmatrix}, \quad (2.56)$$

$$\mathcal{R}_Y^{12}(\theta) = e^{-i\frac{\theta}{2}\lambda_5} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta/2 & \sin \theta/2 \\ 0 & -\sin \theta/2 & \cos \theta/2 \end{pmatrix}, \quad (2.57)$$

$$\mathcal{R}_Z^{12}(\theta) = e^{-i\frac{\theta}{2}\lambda_8^*} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & e^{-i\theta/2} & 0 \\ 0 & 0 & e^{i\theta/2} \end{pmatrix}. \quad (2.58)$$

Now, it can be proven that using this set of 6 unitary matrices, one can obtain universal control over the entire Hilbert space of a single-qutrit. The idea is to factorize an arbitrary $U_3 \in \text{SU}(3)$ matrix into a sequence, or a combination of subspace rotations. This implies that the factorization is not unique. Indeed, based on the specific experimental setup that is readily available, there is a large number of factorization algorithm that results in different sequence of unitaries [6, 27, 22]. Within this thesis, we choose the following sequence,

$$U_3 = \text{diag}(\phi_6, \phi_5, \phi_4) \mathcal{R}_{\phi_3}^{01}(\theta_3) \mathcal{R}_{\phi_2}^{12}(\theta_2) \mathcal{R}_{\phi_1}^{01}(\theta_1), \quad (2.59)$$

where $\text{diag}(\cdot)$ is a diagonal matrix with the coefficients on the diagonal lines; the other rotations are understood as rotation away from the x -axis an angle ϕ and is of the form

$$\mathcal{R}_{\phi}^{01}(\theta) = \begin{pmatrix} \cos \theta/2 & -ie^{i\phi} \sin \theta/2 & 0 \\ -ie^{-i\phi} \sin \theta/2 & \cos \theta/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathcal{R}_Z^{01}(\phi) \mathcal{R}_X^{01}(\theta) \mathcal{R}_Z^{01}(-\phi), \quad (2.60)$$

$$\mathcal{R}_{\phi}^{12}(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta/2 & -ie^{i\phi} \sin \theta/2 \\ 0 & -ie^{-i\phi} \sin \theta/2 & \cos \theta/2 \end{pmatrix} = \mathcal{R}_Z^{12}(\phi) \mathcal{R}_X^{12}(\theta) \mathcal{R}_Z^{12}(-\phi). \quad (2.61)$$

The $\mathcal{R}_Y$ rotations are also subsumed under this formality and is realized by letting $\phi = \pi/2$. The rotations about the x and z axes are therefore more elementary. From now on, these rotations are called elementary rotations. We now give a final remark to our result here,

Remark 5. *Given the access to rotations around the x and z axis, namely $\mathcal{R}_X^S$ and $\mathcal{R}_Z^S$, where (s) defines the two-level subspace (01) and (12) , then universal control over a single-qutrit, i.e. performing an arbitrary unitary operator in the special $\text{SU}(3)$ group is obtained.*

This remark follows directly from (Eqn. 2.59) and (Eqn. 2.60). It is only then left to realize these elementary rotations, namely $\mathcal{R}_X^S$ and $\mathcal{R}_Z^S$ in a real hardware. The way it is done in superconducting qutrit will be introduced formally in Chapter 3. The noise (or error) associated with them is taken into account in Chapter 5.

2.3.3 A toy example: speedup in parity check using qutrit

Although it is beyond the scope of this thesis to give a comprehensive review on how quantum advantage can be obtained using qutrit (see [19, 35]), it is instructive to introduce a minimal example. Here, a quantum ternary algorithm used for determination of the parity of permutation of three elements is presented. Experiment-wise, this has been realized in [16]. This algorithm is one of the simplest algorithm in which we can observe a computational advantage versus its classical counterpart. Let us now consider the simplest case in which we can observe a quantum speedup versus classical version, a set of only three elements $\{1, 2, 3\}$. This set has a total of $3! = 6$ possible permutations. They are $(1, 2, 3) = e$, $(2, 3, 1)$, $(3, 1, 2)$, $(3, 2, 1)$, $(2, 1, 3)$, $(1, 3, 2)$. In Cauchy's two-line notation,

$$\sigma_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \quad \sigma_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \quad \sigma_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \quad (2.62)$$

$$\sigma_4 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \quad \sigma_5 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \quad \sigma_6 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \quad (2.63)$$

As explored above, one way to determine the parity of σ_i is to count its inversions. For a set of m elements, we have to perform at least $(m - 1)$ queries (evaluations) to infer the parity, classically [See Appendix A for proof]. Hence in a set of 3 elements, we have to perform at least 2 evaluations. In quantum case, as we shall see now, we need only a single query.

No speedup case

In a quantum computer, computing the parity of these permutations can be implemented in two ways. One way offers no computational speedup, the other does. Let us look at the no-speedup case first. The quantum oracle is a black box, inside of which, for example, is σ_3 . First we convert σ_3 into a unitary matrix U_σ , such that $U_{ij} = 1$ where $\sigma(i) = j$ and zeroes elsewhere. For example, the unitary representation of σ_3 is

$$U_{\sigma_3} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \quad (2.64)$$

A quick sanity check reveals that $U_{\sigma_i}^\dagger U_{\sigma_i} = I, \forall i$. We might ask what would happen if we apply U_{σ_3} to a basis state of a qutrit, for example $|1\rangle$. The result is a simple transposition of the state vector

$$U_{\sigma_3}|1\rangle = |2\rangle \quad (2.65)$$

indicating that the second element under σ_3 permutation will be sent to the third element. We can repeat this process once again, but this time preparing a different state, for example

$|2\rangle$. The result this time is

$$U_{\sigma_3}|2\rangle = |0\rangle \quad (2.66)$$

indicating that the third element under σ_3 permutation will be sent to the first element. In the first oracle call, we infer that $\sigma_3(2) = 3$. In the second oracle call, we infer that $\sigma_3(3) = 1$. We do not necessarily have to make another oracle call, since we know for sure that $\sigma_3(1) = 2$, since this is a cyclic permutation. Upon making the second measurement, we conclude that there are two inversions, hence the hidden permutation's parity is even (indeed).

Speedup case

We now explore the second case, in which we can infer the parity from a single query. First, we prepare the ground state $|0\rangle$. Next, we apply a quantum Fourier transformation, or as in quantum computation language, a Hadamard gate. This results in the state $H|0\rangle = |\psi_0\rangle$,

$$|\psi_0\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} e^{i\frac{2\pi}{3}} \\ 1 \\ e^{-i\frac{2\pi}{3}} \end{pmatrix} \quad (2.67)$$

in the computational basis $\{|0\rangle, |1\rangle, |2\rangle\}$. Let us now pick an arbitrary permutation σ_k and apply it on the state $|\psi_0\rangle$. The resulting state vector is now $U_{\sigma_k}|\psi_0\rangle \equiv |\psi_k\rangle$, where $|\psi_k\rangle$ written explicitly,

$$|\psi_k\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} e^{i\frac{2\pi}{3}} \\ 1 \\ e^{-i\frac{2\pi}{3}} \end{pmatrix} \quad (2.68)$$

in the basis $\{|\sigma_k(0)\rangle, |\sigma_k(1)\rangle, |\sigma_k(2)\rangle\}$. This state is special, in the sense that

$$|\psi_1\rangle = \exp(-i2\pi/3) |\psi_2\rangle = \exp(i2\pi/3) |\psi_3\rangle \quad (2.69)$$

$$|\psi_4\rangle = \exp(-i2\pi/3) |\psi_5\rangle = \exp(i2\pi/3) |\psi_6\rangle \quad (2.70)$$

This necessarily enforces that, for an oracle containing even permutations (i.e. $\sigma_1, \sigma_2, \sigma_3$), we can only obtain the state $|\psi_0\rangle$. While for an oracle containing odd permutations (i.e. $\sigma_4, \sigma_5, \sigma_6$), we can only obtain the state $|\psi_4\rangle$. We can still obtain the state $|\psi_2\rangle, |\psi_3\rangle, |\psi_5\rangle, |\psi_6\rangle$, but these states remain indistinguishable since they differ $|\psi_0\rangle$ and $|\psi_4\rangle$ up to only a phase factor. We now apply the inverse Fourier transform $H^\dagger|\psi_k\rangle$ and make a single measurement.

- If the parity of σ_k was even, then we detect state $|0\rangle$.
- Otherwise, we detect state $|2\rangle$, indicating that the parity was odd.

Hence, we can determine the parity in a single oracle call.

Circuit quantum electrodynamics

This chapter is intended to give a short overview of the circuit quantum electrodynamics (cQED), the architecture upon which laid the foundation of quantum computing with superconducting circuits. The field took off from a series of seminal work from the Yale's applied physics groups [37, 5, 21, 43]. Utilizing macroscopic objects that exhibit collective quantum mechanical behavior, i.e. the transmon qubit or the 2D/3D cavity, cQED requires a whole new repertoire of tools to process and manipulate the quantum states, and thus opening novel possibilities with μ -wave photon matter interaction based quantum information processing. We would briefly review qubit design (with a focus on transmon qubits), control, and readout techniques utilizing concepts of cQED.

3.1 Introduction

3.1.1 The quantum harmonic oscillator

It is natural to start with the concept of a linear LC circuit. In this system, energy stored in the electric field in the capacitor C and magnetic field within the inductor L . At one instance, the electric energy is

$$E(t) = \int_{-\infty}^t V(\tau)I(\tau)d\tau, \quad (3.1)$$

where $V(\tau)$, $I(\tau)$ are the instantaneous voltage and current of the capacitor and inductor.

It is instructive to view the energy associated with E is the 'kinetic energy', and the energy associated with B is the 'potential energy'. If we picked the generalised coordinates to be the flux Φ , the Lagrangian is

$$\mathcal{L} = \frac{1}{2}C\dot{\Phi}^2 - \frac{1}{2L}\Phi^2, \quad (3.2)$$

from which the Hamiltonian can be found by using the Legendre transformation easily,

$$H = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} = \frac{1}{2}CV^2 + \frac{1}{2}LI^2, \quad (3.3)$$

where the flux is the integration of voltage of the node (which is also at the branch voltage accross the two elements). One can enter the quantum regime using the Dirac formalism where $\hat{\Phi}$ and $\hat{Q}$ are two non-commutative observables satisfying

$$[\hat{\Phi}, \hat{Q}] = i\hbar. \quad (3.4)$$

Let us further define Φ_{ZPF} and Q_{ZPF} such that

$$\Phi_{ZPF} = \sqrt{\hbar/2\omega_r C}, \quad Q_{ZPF} = \hbar\omega_r C/2 \quad (3.5)$$

which represents the characteristics magnitude of the zero-point fluctuations (ZPF) of flux and charge, respectively. This facilitates the introduction of the creation $\hat{a}^\dagger$ (annihilation $\hat{a}$) operators as in any bosonic system,

$$\hat{\Phi} = \Phi_{ZPF}(\hat{a}^\dagger + \hat{a}), \quad (3.6)$$

$$\hat{Q} = iQ_{ZPF}(\hat{a}^\dagger - \hat{a}). \quad (3.7)$$

With these definitions, the Hamiltonian of a LC circuit takes on the usual form,

$$\hat{H} = \hbar\omega_r \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \equiv \hbar\omega_r \hat{a}^\dagger \hat{a}, \quad (3.8)$$

with the eigenstates $|n\rangle$ satisfy $\hat{a}^\dagger \hat{a} |n\rangle = n |n\rangle$, for $n = 0, 1, 2, \dots$. The action of $\hat{a}$ is to create a photon of energy $\hbar\omega_r$ stored in the circuit.

Quantization conditions

It is natural to think about how Dirac's formalism is realized in a physical picture. One cannot simply engineer classical LC-resonators and expect them to behave quantum mechanically, i.e. to be able to store a single quantum excitation. For this to be the case, we have two conditions that must be satisfied. First, the coupling between the oscillator and environment must be extremely small, so as to not introducing and redundant, uncontrolled degrees of freedom. This is realized by embedding superconductor circuits inside dielectric substrates to remedy photon loss rate. Second, the separation of eigen-energies must be larger than the fundamental single-excitation thermal energy. If not, the system is vulnerable to thermal excitation. The condition $\hbar\omega_r \gg k_B T$ can be realized using state-of-the-art dilution refrigerator, with temperature nowadays typically reaching milli-Kelvin scale. At such temperature, a microwave frequency circuit is well-protected from thermal fluctuations, while also far away from superconducting threshold.

Now that we have enter the superconducting quantum regime, it is also helpful to view the Hamiltonian from a different perspective using the reduced flux $\hat{\phi} = 2\pi\hat{\Phi}/\Phi_0$

and reduced charge $\hat{n} = \hat{Q}/2e$, which is the number of the Cooper pairs, and the flux that created when they tunnel through the inductor. The modified version of the Hamiltonian read

$$\hat{H} = 4E_C \hat{n}^2 + \frac{1}{2} E_L \hat{\phi}^2, \quad (3.9)$$

where $E_C = e^2/2C$ is the charging energy required to combine two electrons into a Cooper-pair and add them to the island, and $E_J = (\Phi_0/2\pi)^2/L$ being the rate at which Cooper-pair hopping between two superconducting islands.

3.2 Adding nonlinearities: transmon qubit

The linear properties of the QHO poses a practical limitation. Indeed, quantum information processing requires the ability to address subspaces of the system selectively, i.e. a well-defined computational subspaces. In the QHO case, the equidistant level separation forbids us to manipulate transitions between eigenstates without exciting others by means of control with defined frequency. Therefore, frequency selectivity is a crucial part. We can overcome this by introducing non-linearity to the circuit, the Josephson junction. In summary, the Josephson junction allows us to reshape the potential shape from quadratic to sinusoidal, resulting in non-equidistant energy-spacing structure. Upon replacing the linear inductance L , the current I forms new relationship with the flux ϕ , and can be described in terms of Josephson inductance L_J ,

$$L_J(\Phi) = \left(\frac{\partial I}{\partial \Phi} \right)^{-1} = \frac{\Phi_0}{2\pi I_c} \frac{1}{\cos(2\pi\Phi/\Phi_0)}. \quad (3.10)$$

One can see the resemblance of natural atoms when introducing non-linearity to an LC-oscillator. The energy structure is non-equidistant, and special lines are resolved; hence the common name, *artificial atom*. At the expense of smaller coherence time, we have a larger dipole moment (allowing precise, fast control) and freedom of fabrication and engineering for our special purposes. In light of this, the quantized Hamiltonian can be rewritten as

$$\hat{H} = 4E_C \hat{n}^2 - E_J \cos \hat{\phi}, \quad (3.11)$$

where $\hat{\phi} \equiv (2\pi/\Phi_0)\hat{\Phi}$. The electric energy is the total energy stored in two capacitors C_s (shunted) and C_J (Josephson), which is $e^2/2(C_s + C_J)$. The magnetic energy is stored in the field around the critical current, $E_J = I_c \Phi_0/2\pi$, also known as Josephson energy. Once the non-linearity has been introduced, the Hamiltonian is now completely described by the ratio of E_J/E_C . The majority of qubit design these days employs $E_J \gg E_C$ so as to gain robustness against charge noise. Typically qubits are now fabricated with the ratio $E_J/E_C \approx 50 - 70$, this is called the *transmon regime* [21]. A prominent nature of artificial atom is however not being captured in this Hamiltonian: static coupling of the

charge degree of freedom to environmental electric drifts. For linear LC oscillator, this static drift can be removed by a canonical transformation of coordinates (either φ or n); this is also reflected in the equidistant spacing of the energy levels—it is then harder for a malicious environment to disturb the system). For nonlinear LC oscillator, for example the artificial atom that we consider here, this static drift has to be included to the Hamiltonian through a static offset number, namely n_g (the "g" letter represents gate electrode, for convention). The Hamiltonian is therefore

$$\hat{H} = 4E_C(\hat{n} - n_g)^2 - E_J \cos \hat{\varphi}, \quad (3.12)$$

where n_g is measured in the units of the Cooper pair charge. Note that Hamiltonian is periodic, $H(\varphi + 2\pi) = H(\varphi)$ and therefore the stationary Schrodinger equation can be solved analytically in the phase basis using Mathieu's functions (See [5] or [21]). As for the rest of this thesis, working the charge basis results in a clearer physical picture: it both captures the so-called charge noise that is largely presented in real hardware and generally easier to simulate using a laptop.

3.2.1 Charge dispersion

It is preferred to express the Hamiltonian in (Eqn. 3.12) in the charge basis $|n\rangle$ which are the eigenstates of the number operator that we introduced previously, $\hat{n}|n\rangle = n|n\rangle$. In a truncated charge basis $N_{max} = N$ (meaning we dealing with a finite dimension Fock space) [5], the Hamiltonian is

$$H = 4E_C \sum_{j=-N}^N (j - n_g)^2 |j\rangle\langle j| - E_J \sum_{j=-N}^{N-1} (|j+1\rangle\langle j| + |j\rangle\langle j+1|). \quad (3.13)$$

This Hamiltonian can be easily diagonalized using numerical methods. Plots the eigenenergies of this Hamiltonian, as shown in Fig. 3.3, as a function of charge offset n_g . It is then evident that the system's dependence on environmental fluctuation coupled to the charge's degree of freedom is now exponentially suppressed. To further illustrate this point, we calculate the so-called *charge dispersion* of the transmon in different E_J/E_C regime. Denotes ϵ_m the peak-to-peak energy range of the m -level, then

$$\epsilon_m = E_m(n_g = 1/2) - E_m(n_g = 0). \quad (3.14)$$

In the limit of small charge dispersion ($E_J/E_C \rightarrow 20$, tentatively), the dispersion relation is approximated by a cosine relation,

$$E_m(n_g) \approx E_m(n_g = 1/4) - \frac{\epsilon_m}{2} \cos(2\pi n_g). \quad (3.15)$$

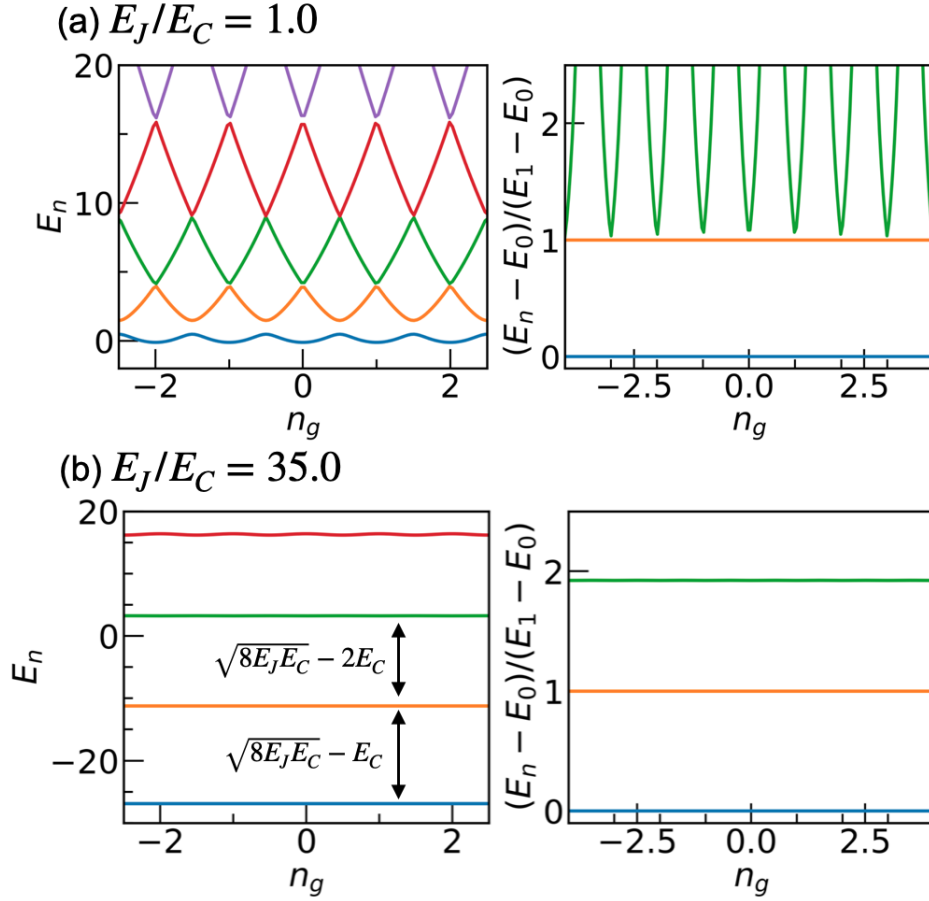


Figure 3.1: Energy structure of the Hamiltonian given in Eqn. 3.13. (a) Five lowest energy levels of the transmon ($E_J/E_C = 1$); this structure exhibits clear dependence of the energies on offset charge n_g with avoided crossings. (b) Four lowest energy levels of the transmon ($E_J/E_C = 35.0$). For the first three energy levels, the dependence is exponentially suppressed.

As pointed out above, the eigenenergies E_m can be analyzed in the phase basis using Mathieu's function (see [21]). Using this, we obtain the charge dispersion dispersion,

$$\epsilon_m \approx (-1)^m \frac{E_C 2^{4m+5}}{m!} \sqrt{\frac{2}{\pi}} \quad (3.16)$$

$$\times (E_J/2E_C)^{2m+3/4} e^{-\sqrt{8E_J/E_C}}, \quad (3.17)$$

which tends to $\epsilon_m \rightarrow 0$ as $E_J/E_C \rightarrow \infty$, which also means the anharmonic oscillator tends to a harmonic oscillator, i.e. the anharmonicity goes to zero. It is however clear that while ϵ_m decreases much faster (exponential law) compared to the system's anharmonicity. Hence, a large E_J/E_C that yields smaller charge dispersion is favoured since it translates to insensitivity with charge noise.

3.2.2 Anharmonicity of the transmon

To make $E_J \gg E_C$, the preferred way is to make the capacitance of the Josephson junction to be smaller than that of the shunted. In this limit, the superconducting phase

$\hat{\varphi}$ is small (i.e. $\langle \varphi \rangle^2 \rightarrow \infty$)¹. Let us rewrite the Hamiltonian by means of the Maclaurint series,

$$\hat{H} = 4E_C \hat{n}^2 + \frac{1}{2} E_J \hat{\varphi}^2 - \frac{1}{4!} E_J \hat{\varphi}^4. \quad (3.18)$$

The non-quadratic term of the Josephson energy goes as $1/4!$, thus implying the transmon a weakly-anharmonic oscillator. Following the same argument above, it is useful to look at $\hat{H}$ from the bosonic picture by denoting

$$\hat{\varphi} = \left(\frac{2E_C}{E_J} \right)^{1/4} (\hat{b}^\dagger + \hat{b}), \quad (3.19)$$

$$\hat{n} = \frac{i}{2} \left(\frac{E_J}{2E_C} \right)^{1/4} (\hat{b}^\dagger - \hat{b}). \quad (3.20)$$

The Hamiltonian is now transform in to annihilation (creation) basis,

$$\hat{H} = \sqrt{8E_C E_J} \hat{b}^\dagger \hat{b} - \frac{E_C}{12} (\hat{b}^\dagger + \hat{b})^4 \quad (3.21)$$

$$\approx \hbar \omega_b \hat{b}^\dagger \hat{b} - \frac{E_C}{2} \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{b}, \quad (3.22)$$

where ω_b is the plasma frequency, $\sqrt{8E_C E_J} - EC$. We have also used the quartic expansion approximation, $(\hat{b}^\dagger + \hat{b})^4$ by removing terms that do not conserve excitation number. The non-linearity term $\alpha/\hbar = E_C/\hbar$ denotes the *anharmonicity* of the system, i.e. the difference between the transition frequency of the first two subspace $\omega_{12} - \omega_{01} \equiv \alpha$. Typically for a transmon $\alpha \approx 100 - 400$ MHz [36, 21, 6, 10, 45]. Note that this value is highly related to the so-called "sweet-spot"; further increase of $E_J/E_C \rightarrow \infty$ results in an increasingly smaller system's anharmonicity $\alpha \rightarrow 0$. In other words, we asymptote the quantum harmonic oscillator limit. If leakage to higher subspace is suppressed, we can treat QHOs as a spin-1/2 system,

$$\hat{H} = \omega_q (\hat{\sigma}_z/2) \quad (3.23)$$

However, one should always keep in mind that higher levels exist. In fact, the quest to mitigate leakage out of computational subspace is an active area of research up to this date. A convincing picture would be the Fourier transform of the microwave field used to manipulate the state of the transmons results in multiple unwanted components that will induce the so-called "phase-error" and "leakage" to the computational subspace.

It is also useful to recast the Hamiltonian 3.21 into the charge basis again. This can

¹Or in other words, the charge degree of freedom $\hat{n}$ is further de-localized.

be done by mapping the bosonic ladder operators into number operators

$$\hat{b} = \sum_j n_{j,j+1} |j\rangle \langle j+1|, \quad (3.24)$$

$$\hat{b}^\dagger = \sum_j n_{j+1,j} |j+1\rangle \langle j|, \quad (3.25)$$

where $n_{i,j}$ are the coefficients of the truncated bosonic operator.

3.2.3 Coupling to cavity

Just as when we place an atom inside a Fabry-Perot cavity, it is also possible to place the artificial transmon atom in a cavity. This is done by direct dipole coupling $\hat{n} \propto (\hat{b}^\dagger + \hat{b})$ (or in electrical circuit language, capacitive coupling) between the dipole of the atom and one dominant mode of the electric field. The dipole coupling strength, denotes g , is of a different order of magnitude compared to the usual coupling strength obtained within the regime of cavity quantum electrodynamics. The full Hamiltonian in this case, containing terms related to the transmon, cavity, and their interaction is

$$\hat{H} = 4E_C(\hat{n} - n)^2 - E_J \cos \hat{\varphi} + \hbar\omega_c \hat{a}^\dagger \hat{a} + \hbar\tilde{g}\hat{n}(\hat{a} + \hat{a}). \quad (3.26)$$

It is instructive to move the transmon back to the bosonic picture and using the approximated Hamiltonian derived in Eqn. 3.21,

$$\hat{H} = \hbar\omega_b \hat{b}^\dagger \hat{b} - \frac{\alpha}{2} \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{b} + \hbar\omega_a \hat{a}^\dagger \hat{a} + \hbar g (\hat{b}^\dagger + \hat{b})(\hat{a} + \hat{a}), \quad (3.27)$$

where ω_b is the plasma frequency and $\alpha/\hbar$ is the anharmonicity as mentioned. For simplicity, we assume that the transmon is strongly coupled to exactly one mode of the cavity (while the other modes are far detuned from it). The optical frequency of the mode is ω_a . The coupling strength, g in this case, can be estimated as

$$\hbar g = -\sqrt{2}ieV_a(8E_C/E_J)^{-1/4}, \quad (3.28)$$

where V_a is the strength of effective voltage fluctuations due to the electric field, E_C/E_J ratio is controlled by the transmon. The fact that g is allowed to be complex is by convention [5]. The Equation 3.27 can be further simplified in the experimentally relevant situation where the coupling constant is small compared to other system's frequency modes, namely $|g| \ll \omega_b, \omega_a$. Using the rotating-wave approximation,

$$\hat{H} = \hbar\omega_b \hat{b}^\dagger \hat{b} - \frac{\alpha}{2} \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{b} + \hbar\omega_a \hat{a}^\dagger \hat{a} + \hbar g (\hat{b}^\dagger \hat{a} + \hat{b} \hat{a}^\dagger), \quad (3.29)$$

the Hamiltonian represents a Jaynes-Cummings-like dynamics. This is more appreciated when making the replacements $\hat{b} \rightarrow \hat{\sigma}_+ = |1\rangle\langle 0|$ and $\hat{b} \rightarrow \hat{\sigma}_- = |0\rangle\langle 1|$, whom are the

raising/lowering operator in a two-level atom, then

$$\hat{H}_{JC} = \hbar\omega_a \hat{a}^\dagger \hat{a} + \hbar(\omega_b/2)\hat{\sigma}_z + \hbar g(\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+). \quad (3.30)$$

When ω_a is close to ω_b , the last interaction term of Eqn. 3.30 describes a coherent exchange of a single quantization of energy between the electric field and the atom: either the cavity receives a photon and the atom is de-excited or reverse. When $\omega_a = \omega_b$, the system is maximally entangled, meaning that the reduced density matrix of the atom (or the transmon qubit) traced out the cavity is not a pure state. This is not desired since it means we *do not* have an isolated qubit. As should be clear later, this resonance regime is often replaced by the dispersive regime, where the qubit-cavity detuning is large compared to the coupling strength, $|g/\Delta| \ll 1$.

3.2.4 Encoding a qutrit in transmon

It should be always noted that the transmon qubit is a multi-level system. To realize a qutrit, the simplest idea is to confined the computational subspace to the lowest subspace spanned by the first three eigenstates of the transmon's Hamiltonian. That is, given Eqn. 3.21, the first three eigenstates being the $|0\rangle, |1\rangle, |2\rangle$ eigenstates of the $\hat{n}$ numer operator itself (since obviously $[\hat{n}, \hat{H}] = 0$ and the Hamiltonian is diagonal in this basis). However, in practice we will also have to take into account the qubit-cavity coupling g_{ab} and qubit-drive coupling Ω . This will complicate the scheme since there will be energy reordering inside the system and we always therefore work with the dressed state of the system $|\bar{\psi}\rangle \neq |\psi_{\text{qutrit}}\rangle \otimes |\psi_{\text{cav}}\rangle \otimes |\psi_{\text{drive}}\rangle$. To overcome this problem, as prompted, we move into the dispersive regime to further simplify the qubit-cavity interaction. Moreover, we also assume that the driving terms are within the rotating-wave approximation and can be captured perturbatively. That is, once the pulse ramped up and driven the transmon states becomes mixed with the EM field, but after ramping down the states return to their uncoupled version. In fact, what we can measure spectroscopically is always the weakly coupled states of the qubit-cavity system. However, by moving to the dispersive regime, we can safely assumed that $|\bar{\psi}\rangle \approx |\psi_{\text{qutrit}}\rangle \otimes |\psi_{\text{cav}}\rangle$; that is, albeit being entangled, we have an *approximated* isolated qubit.

3.3 Driving a transmon

3.3.1 Introducing the drive

A quantum system can be controlled by a classical drive. The drive, interchangeably called as a pulse, is assumed to be monochromatic and normalized, that is

$$\int_0^T s(t)dt = 1, \quad (3.31)$$

where $s(t)$ is the envelope of the pulse. The monochromatic condition is guaranteed when $T \gg \omega^{-1}$, where ω is the optical frequency of the pulse. The classical drive is then quantized and assumed to be a single-mode field, $\omega_d \hat{d}^\dagger \hat{d}$. The dipole interaction between the drive and the transmon is

$$\hat{H}_{\text{int}} = -\vec{E}(t) \cdot \hat{n}, \quad (3.32)$$

$$= \tilde{V}_d(t) \cos(\omega_d t + \theta_d) \cdot i(8E_J/E_C)^{1/4}(b - b^\dagger)\sqrt{2}, \quad (3.33)$$

$$= \mathcal{E}_d(t) \cos(\omega_d t + \theta_d)(b - b^\dagger), \quad (3.34)$$

where the constants are absorbed into the pulse envelope; where can define $\mathcal{E}_d(t) = \mathcal{E}_0 s(t)$ where E_0 is commonly referred to as the Rabi driving strength, $s(t)$ is the dimensionless the pulse envelope and is normalized, and the phase of the classical drive is θ_d . An important feature of this pulse is that, it can be recasted in the form of two quadrature components 90 degrees out of phase to each other, $\mathcal{E}_x, \mathcal{E}_y$,

$$\mathcal{E}_d(t) = \mathcal{E}_x(t) \cos(\omega_d t + \theta_0) + \mathcal{E}_y(t) \sin(\omega_d t + \theta_0). \quad (3.35)$$

where the θ_0 phase of the two components vanishes when making a rotating-wave approximation. Using simple trigonometry, the pulse is rewritten as

$$\mathcal{E}_d(t) = \frac{1}{2} [(\mathcal{E}_x(t) + i\mathcal{E}_y(t))e^{-i(\omega_d t + \theta_d)} + (\mathcal{E}_x(t) - i\mathcal{E}_y(t))e^{i(\omega_d t + \theta_d)}], \quad (3.36)$$

$$= \frac{1}{2} [\mathcal{E}(t)e^{-i(\omega_d t + \theta_d)} + \mathcal{E}^*(t)e^{i(\omega_d t + \theta_d)}]. \quad (3.37)$$

where $\mathcal{E} = \mathcal{E}_x + i\mathcal{E}_y$. Within the rotating-wave approximation, when transforming into the frame of the drive using the unitary transformation $U = e^{-i\omega_d t \hat{b}^\dagger \hat{b}}$, the interaction Hamiltonian reduces to

$$\hat{H}_{\text{int}} = \mathcal{E}^*(t)\hat{b} + \mathcal{E}(t)\hat{b}^\dagger. \quad (3.38)$$

3.3.2 Single-qutrit gates: realization

Consider a standard nonlinear oscillator (SNO) [15, 28], where the ground and excited state manifolds are separated by $\hbar\omega$. The j -th energy levels are different to $\hbar j\omega$ (that of a harmonic oscillator) by $\hbar\Delta_j$, also known as the anharmonicity. The Hamiltonian for a d -dimension SNO is

$$H_0 = \sum_{j=1}^{d-1} (j\omega - \Delta_j) |j\rangle\langle j|, \quad (3.39)$$

where $|j\rangle\langle j| = P_j$ is exactly the rank-1 projector onto the j -th energy level subspace. By definition, we have $\Delta_1 = 0$, and by convention the anharmonicity $\Delta_j = \Delta_2(j-1)j/2$. Since we have assume that control of this SNO is due to the dipole interaction between

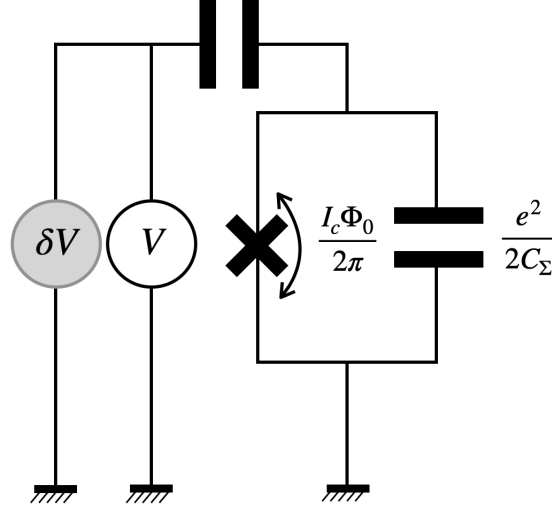


Figure 3.2: Coupling of the transmon to a drive. In this lumped picture, the transmon is a circuit capacitively shunted by a Josephson junction. The energy stored in the capacitor is $e^2/2C_\sigma$, where C_σ is the total effective capacitance of the circuit. The energy stored in the Josephson junction is $I_c\Phi_0/2\pi$, which is also the rate at which a charge particle jumping between the two superconducting islands. The microwave driveline is characterized by a time-dependent voltage $V(t)$. The effect of noise is represented by an always on offset drive line δV .

the oscillator and the electric field, the driving (interaction) Hamiltonian reads,

$$H_d(t) = \mathcal{E}_d(t) \sum_{j=1}^{d-1} \lambda_{j-1} (|j\rangle\langle k| + |k\rangle\langle j|), \quad (3.40)$$

where $\lambda_0 = 1$ (i.e. the strength of transition between $|0\rangle$ and $|1\rangle$) and the rest are weighted relative transition strength between the $|j\rangle$ and $|j+1\rangle$ states. As with the bosonic picture, $\lim_{d \rightarrow \infty} |j\rangle\langle k| + |k\rangle\langle j| \rightarrow \hat{b} + \hat{b}^\dagger$. The weighted transition strength, to a good approximation, follows that of the harmonic oscillator $\lambda_j = \sqrt{j}$; however, it is often pointed out that this approximation does not hold in various case and has been experimentally verified (See [28, 15] for theory, [44] for experiment). Now, we go into the frame rotating at the drive frequency ω_d again, by making unitary transformation $U(t) = \sum_{j=1}^{d-1} \exp(-i(j\omega_d t)|j\rangle\langle j|)$,

$$H'(t) = U^\dagger(t)H(t)U(t) + i\dot{U}^\dagger(t)U(t), \quad (3.41)$$

where $H(t) = H_0 + H_d(t)$ is the total Hamiltonian describing the SNO and its interaction with the field. Explicitly we have

$$H'(t) = \sum_{j=1}^{d-1} (j\delta + \Delta_j) |j\rangle\langle j| + \left[\frac{\mathcal{E}(t)}{2} e^{-i(\omega_d t + \theta_d)} + \text{h.c.} \right] \times \left[\sum_{j=1}^{d-1} \lambda_{j-1} (|j\rangle\langle j-1| e^{i\omega_d t} + \text{h.c.}) \right], \quad (3.42)$$

where h.c. stands for Hermitian conjugate and the pulse envelope definition is borrowed from Eqn. 3.36. In this frame, we can throw away terms that oscillate twice as fast as the drive frequency $2\omega_d$ (rotating-wave approximation). Then, the Hamiltonian reads,

$$H'(t) = \sum_{j=1}^{d-1} (j\delta - \Delta_j) |j\rangle\langle j| + \sum_{j=1}^{d-1} \lambda_{j-1} \left[\frac{\mathcal{E}(t)}{2} |j\rangle\langle j-1| + \frac{\mathcal{E}^*(t)}{2} |j-1\rangle\langle j| \right]. \quad (3.43)$$

Connection to the Gell-Mann basis

To avoid notation clash, where the weighted transition strength λ is confused with the Gell-Mann basis λ_j , we now suppose that $\lambda_1 = \sqrt{2}$ (in addition to the convention $\lambda_0 = 1$, taken from the harmonic oscillator). Now, if we restrict ourselves to only the first three levels, namely $d = 3$, then

$$\begin{aligned} H'(t) = & \delta |1\rangle\langle 1| + (\delta - \Delta_2) |2\rangle\langle 2| + \frac{1}{2} [\mathcal{E}(t) |1\rangle\langle 0| + \mathcal{E}^*(t) |0\rangle\langle 1|] \\ & \times \sqrt{2} \frac{1}{2} [\mathcal{E}(t) |2\rangle\langle 1| + \mathcal{E}^*(t) |1\rangle\langle 2|], \end{aligned} \quad (3.44)$$

or written in the basis of $|j\rangle$,

$$H'(t) = \begin{pmatrix} 0 & \frac{1}{2} (\mathcal{E}_x(t) + i\mathcal{E}_y(t)) & 0 \\ \frac{1}{2} (\mathcal{E}_x(t) - i\mathcal{E}_y(t)) & \delta & \sqrt{2} \frac{1}{2} (\mathcal{E}_x(t) + i\mathcal{E}_y(t)) \\ 0 & \sqrt{2} \frac{1}{2} (\mathcal{E}_x(t) - i\mathcal{E}_y(t)) & \delta - \Delta_2 \end{pmatrix}, \quad (3.45)$$

$$= \text{diag}(0, \delta, \delta - \Delta_2) + \frac{\mathcal{E}_x(t)}{2} \lambda_1 + \frac{\mathcal{E}_y(t)}{2} \lambda_2 + \frac{\mathcal{E}_x(t)}{\sqrt{2}} \lambda_4 + \frac{\mathcal{E}_y(t)}{\sqrt{2}} \lambda_5, \quad (3.46)$$

where λ_j are the Gell-Mann matrices. They describe the rotation within the subspaces spanned by $\{|0\rangle, |1\rangle\}$ and $\{|1\rangle, |2\rangle\}$. If we turn on the EM field for a time t_g , we effectively realize a unitary

$$U = \mathcal{T} \exp \left[-i \int_0^{t_g} H'(t) dt \right], \quad (3.47)$$

where t_g is also called gate time, $\mathcal{T}$ is the time ordering operator. By judicious choice of the drive amplitude and frequency, we can effectively realize the rotations specified in Eqn. 2.59. For example, when (1) $\omega_d = \omega$, i.e. on resonance with the transition frequency of the ground and excited manifolds and (2) for simplicity let $\mathcal{E}_y(t) = 0$; then $\delta = 0$ and the effective unitary is

$$U_{\text{eff}} = e^{i\phi} \left[- \left(i \int_0^{t_g} \frac{\mathcal{E}_x(t)}{2} dt \right) \lambda_1 \right] \oplus U_{\text{rest}}. \quad (3.48)$$

If we turn on the field for an specific $t_g = \tau$ such that $\int_0^\tau \mathcal{E}_x(t) dt = \pi$, then

$$U_{\text{eff}} = e^{i\phi} \mathcal{R}_X^{01}(\pi) \oplus U_{\text{rest}}, \quad (3.49)$$

where U_{rest} is some extra terms whose effect should be removable by some other methods. This is one of the central results of this thesis. This point will be illustrated further along this thesis in Chapter 4 where the phase advance problem is introduced and in Chapter 5 where the theory of Derivative Removal of Adiabaticity (or DRAG) is introduced.

3.4 Readout transmon

Measurement in quantum computing is inspired by the ideal of David Mermin, *be silent and calculate*. So far, the statistical data agrees excellently with theoretical prediction; thus, we have a strong motivation to sweep the ‘problem’ under the rug and focus on engineering better quantum systems. With this in mind, let us now consider the dispersive readout technique which is widely used nowadays in circuit QED architecture.

3.4.1 Dispersive regime

As noted above, we want the reduced density matrix of the transmon qubit to be approximately pure when traced out the cavity’s degree of freedom. This is achieved when we move into the dispersive regime. Physically, this corresponds to the situation when the cavity and transmon are sufficiently detuned compared to their coupling strength, $|g|/\Delta \ll 1$. Using the transformation defined by the following unitary

$$U = \exp \left[\sum_j \lambda_j |j\rangle \langle j+1| a^\dagger - \text{h.c.} \right], \quad (3.50)$$

then to the second order within the rotating-wave approximation, the Hamiltonian describing the coupled transmon-cavity system (Eqn. 3.27) then becomes

$$\tilde{H} \approx \hbar\omega_a a^\dagger a + \hbar\omega_b b^\dagger b - \frac{\alpha}{2} b^\dagger b^\dagger b b \quad (3.51)$$

$$+ \sum_{j=0}^{\infty} \hbar(\Lambda_j + \chi_j a^\dagger a) |j\rangle \langle j|, \quad (3.52)$$

where $|j\rangle$ label the eigenstates of the transmon as before. We have also defined $\Lambda_j = \chi_{j-1,j}$ and χ_j are famously known as the dispersive shifts, while Λ_j are known as the Lamb shifts. If we confine the transmon to the first lowest excitation manifold, $|g\rangle, |e\rangle$ then the Hamiltonian is approximated as

$$H_{\text{disp}} = \hbar\tilde{\omega}_a + \frac{\hbar\tilde{\omega}_b}{2} \sigma_z + \hbar\chi a^\dagger a \sigma_z, \quad (3.53)$$

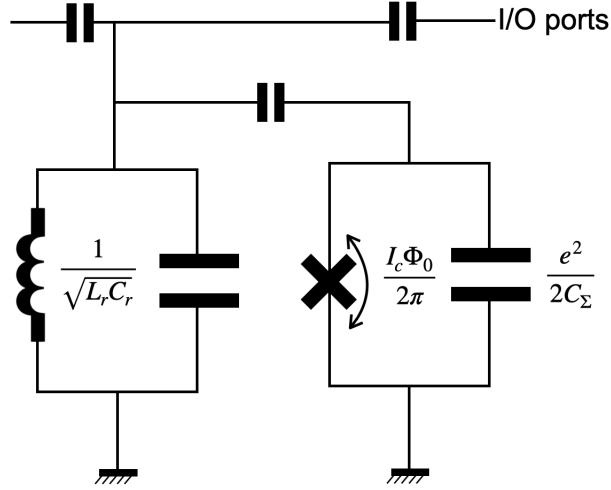


Figure 3.3: Transmon readout. The transmon is capacitively coupled to a cavity/resonator for readout. In the lumped picture, the resonator is a quantized LC circuit with the fundamental frequency $\omega_r = 1/\sqrt{L_r C_r}$.

where χ is the transmon state-dependent dispersive cavity-shift with

$$\omega'_a = \omega_a - \frac{g^2}{\Delta - \alpha/\hbar}, \quad (3.54)$$

$$\omega'_b = \omega_b + \frac{g^2}{\Delta}, \quad (3.55)$$

$$\chi = -\frac{g^2 \alpha / \hbar}{\Delta(\Delta - \alpha/\hbar)}. \quad (3.56)$$

These three dressed frequencies are what being routinely measured in experimental settings. The results shifted frequency of the transmon will be provided in Chapter 4.

3.4.2 Dispersive readout: the non-demolition aspect

The most prominent feature of the dispersive Hamiltonian (Eqn. 3.54) is that the frequency of the cavity is conditioned on the state of the qubit. This fact is being used to a great extent for the readout of the state of the qubit.

If the qubit, for example, is in $|g\rangle$ then $\langle \hat{\sigma}_z \rangle = -1$, and the resonator is shifted to the left $\omega_r - \chi$; if the qubit is in $|e\rangle$, the frequency is shifted to the right $\omega_r + \chi$. Driving the cavity in this situation also induce non-resonant drive on the qubit, since the pair is now evolving as an entangled qubit-resonator state. If, the qubit is in a superposition $c_g|g\rangle + c_e|e\rangle$, then the total wave function is

$$c_g|g, \alpha_g\rangle + c_e|e, \alpha_e\rangle. \quad (3.57)$$

One can think of the Stern-Gerlach experiment for a good depiction. As the atom passes through the magnetic field, a spin-dependent force is applied so that the spin state is entangled with the momentum state. We usually talk about the spin of the atom, but in

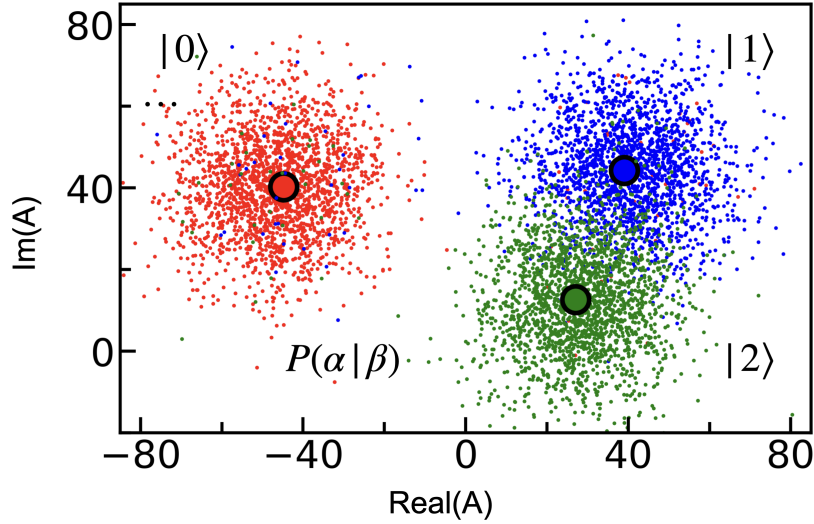


Figure 3.4: The IQ plane. Reflected signals from a resonator conditioned on the qutrit's state. Each cluster corresponds to the bare eigenstates of the transmon, $|0\rangle$ (red), $|1\rangle$ (blue), $|2\rangle$ (green). The $|1\rangle$ (blue) and $|2\rangle$ (green) clusters are not resolved with high precision, prompting that the shifted frequency is not as large as the shifted frequency between the $|0\rangle$ (red) and $|1\rangle$ (blue).

fact what we have measured is the position of it, i.e. on which side of the detector the atom landed. By analogy, the spin-dependent state of the microwave field $\alpha_{e,g}$ can be measured in a heterodyne detector, which allows us to interrogate the qubit state. On the other hand, the frequency of the qubit is also shifted an amount of g^2/Δ which called a Lamb shift. Since we are in the realm of quantum optics, i.e. small photon-number $\hat{a}^\dagger \hat{a}$, the interaction term commutes with $\hat{\sigma}_z$; therefore allowing a quantum-non demolition measurement (QND).

3.4.3 Qutrit readout on transmon

Let us now return to the general dispersive Hamiltonian, c.f. Eqn. 3.51,

$$\tilde{H} = \sum_j (\omega_j + \chi_{j-1,j} + \chi_j a^\dagger a) |j\rangle \langle j| + \omega_r a^\dagger a, \quad (3.58)$$

where we have replaced a with r , denoting the resonator and b with j , denoting each energy-eigenstates of the qudit. Using the same idea presented in the qubit case, where the frequency of the cavity experiences a "pull" conditioned on the transmon's state, we can send a signal of known phase and amplitude,

$$E_{\text{in}}(t) = A_{\text{in}} \cos(\omega_{\omega_d} t + \phi_{\text{in}}), \quad (3.59)$$

and make measurement on the reflected signals

$$E_{\text{out}}(t) = A_{\text{out}} \cos(\omega_{\omega_d} t + \phi_{\text{out}}). \quad (3.60)$$

On IBM Quantum hardware, the reflected signals are coherent state of a low-Q resonator capacitively coupled to the transmon. This low-Q cavity has a lifetime of several hundreds nanoseconds, at most; therefore when we send the signal in it rapidly leaks out the state. This state, represented by a mean-field amplitude A , when integrated gives one a complex number, $z = x + iy$,

$$\bar{A} = \int_0^T dt k(t) A, \quad (3.61)$$

where $k(t)$ is the kernel integration instructions and T is the total duration of them measurement. The measurement is effectively a 50/50 heterodyne detection. Repeating the measurement N times for each preparation of the $|0\rangle, |1\rangle, |2\rangle$ states yields the so-called IQ plane, see Fig. 3.4 that will be used for discriminate different qutrit states in subsequent experiments.

Characterization of a transmon qutrit

In this chapter, we describe how to calibrate the control of a single superconducting transmon qutrit. The procedure commences with the measurement of the Lamb-shifted frequency of the manifolds $\{|0\rangle, |1\rangle\}$, and $\{|1\rangle, |2\rangle\}$, namely ω_{01} and ω_{12} . Having these transitions frequency, we turn to the problem of finding the Rabi driving strength $\mathcal{E}_0$ by observing the Rabi oscillation on each subspace. Next, the transition frequencies are revisited. By performing the Ramsey experiments, we can push the limits of measuring ω_{01} and ω_{12} as compared to being done spectroscopically. The spectral density reveals a major drawback of realizing qutrit (or more generally, qudit) on IBM Quantum Hardware due to its increased sensitivity to charge noise on higher level states. In all experiments described from here, we used a single-junction Josephson quantum processor made cloud-accessible by the IBM Quantum Experience program [2]. The qutrit is engineered from a 127-qubit processor named IBM-BRISBANE with transversal relaxation time $T_1 \approx 200 \mu\text{s}$ and Hahn-echoed dephasing time $T_2^E = 50 \text{ ns}$. The $\pi/2$ rotations about the x axis in the (01) and (12) subspace are chosen to be the elementary physical gates and are realized by resonant microwave pulses with dimensionless Gaussian envelopes. The amplitude of these Gaussian pulses are chosen such that they effectively drive a quarter period Rabi oscillation. Rotations about the z -axis are carried out by shifting the phase of the subsequent pulses [26] and assumed error-free. The transmon's anharmonicity separates resonant frequencies $f_{01} = 4.9 \text{ GHz}$ and $f_{12} = 4.5 \text{ GHz}$ by an amount of $\Delta/2\pi = -340 \text{ MHz}$. From perturbation theory, the estimated E_J/E_C ratio of our transmon qutrit is approximately 37, rendering the (12) subspace sensitive to offset charge fluctuations.

4.1 Qutrit spectroscopy

The scope of this section is to measure the transition frequencies of the $\{|0\rangle, |1\rangle\}$, and $\{|1\rangle, |2\rangle\}$ subspace, denoted as ω_{01} and ω_{12} . The protocol to measure these two parameters is

1. Prepare the qutrit in its ground state $|0\rangle$.

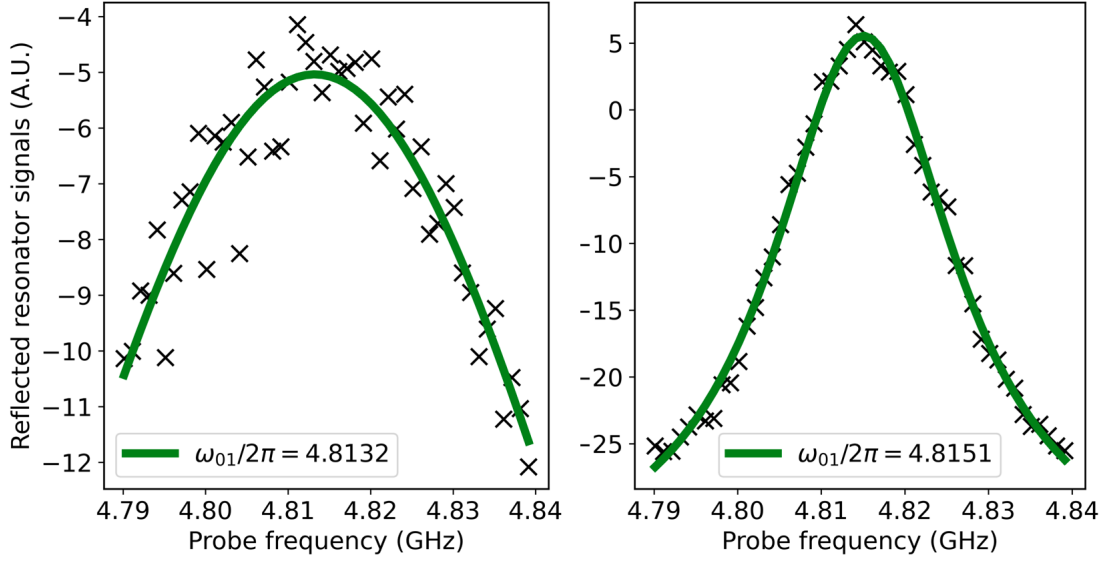


Figure 4.1: Spectroscopy results of ω_{01} . (Left) Spectroscopy results done using a large bandwidth pulse (i.e. short pulse). The pulse duration used was 16 ns. The FWHM is wide, indicating a small signal to noise ratio. (Right) Spectroscopy results done using a small bandwidth pulse (i.e. long pulse). The pulse duration used was 60 ns. The curve clearly is a Lorentzian curve with non-resolving peak.

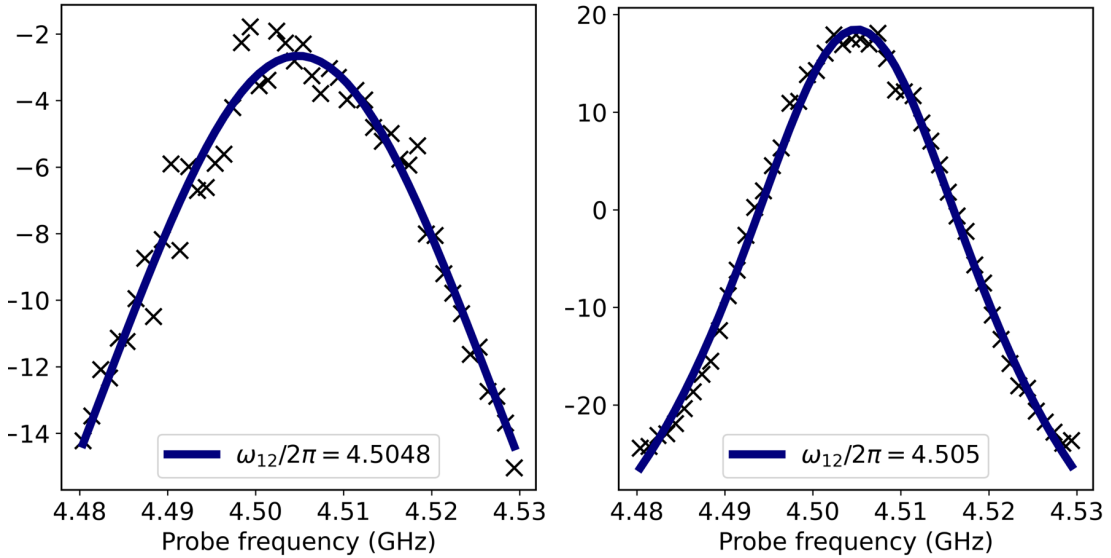


Figure 4.2: Spectroscopy results of ω_{12} . (Left) Spectroscopy results done using a large bandwidth pulse (i.e. short pulse). The pulse duration used was 16 ns. The FWHM is wide, indicating a small signal to noise ratio. (Right) Spectroscopy results done using a small bandwidth pulse (i.e. long pulse). The pulse duration used was 60 ns. The curve follows that of a Lorentzian, but shows deviation compared to the (01) subspace.

2. Apply a Gaussian spectroscopy tone, also known as a probe, with frequency varied in range ± 50 MHz around the suspected value. The tone's amplitude is freely chosen. The duration must be chosen judiciously.
3. Apply a measurement tone on the resonator.
4. Measuring the reflected signals on the resonator using a heterodyne detection scheme.
5. Fit the obtained IQ data set. Assuming a Lorentzian-type of function.

In a superconducting system, a ground state can be actively or passively prepared. Due to bosonic enhancement, higher energy states quickly decay to the ground state when turned on the system. By waiting until the system reaches its quasi-equilibrium state, the qubit's ground state $|0\rangle$ can be prepared with $F_{\text{assign}} \approx 99.9\%$ [1]. For the $\omega_{12}/2\pi$ parameter, we add one π^{01} pulse at the beginning of the circuit to prepare $|1\rangle$. The spectroscopy tone is also an electromagnetic pulse. A Gaussian pulse characterized by three parameters, namely amplitude, frequency, and duration. The amplitude is kept to an arbitrarily low number (since $0 \leq s(t) < 1$). It is however disproportional to the duration. The duration, as should be clear in Figure 4.1 clearly affects the FWHM of the measured signals. For this reason, a probe duration of 16 ns requires 0.3 Rabi driving strength, while a probe duration of 60 ns requires only 0.1 Rabi driving strength. The measurement tone on the resonator is calibrated by IBM's internal team. The Lorentzian-type of function reads,

$$g(f) = \frac{A}{\pi} \frac{B}{(f - f_0)^2 + B^2} + C, \quad (4.1)$$

where A, B, C are free parameters and f_0 is the parameter of interest. For our system, on qubit 1 the spectroscopically measured transition frequencies are $\omega_{01}/2\pi = 4.8151$ GHz and $\omega_{12}/2\pi = 4.5051$ GHz.

4.2 Rabi oscillation

Having determined the transition frequencies of the (01) and (12) subspace. We now turn to the protocol finding the exact envelope's amplitude/duration for realizing single qutrit gates. To this end, we use the Rabi oscillation phenomenon on the transmon qutrit. Rabi oscillations can be understood in light of the Hamiltonian presented above (Eqn. 3.32),

$$H(t) = -\delta_{db}b^\dagger b - \mathcal{E}_d(t) \cos(\omega_d t + \theta_d)(b - b^\dagger), \quad (4.2)$$

$$= -\delta_{db}\sigma_z - \mathcal{E}_d(t) \cos(\omega_d t + \theta_d)\sigma_x. \quad (4.3)$$

Where we have used the two-level approximation $b^\dagger \rightarrow \sigma$. Upon resonance $\delta_{db} = 0$, the microwave field induces coherent transitions between the two computational states of the σ_z that can be measured by coupling the transmon to the resonator. If, for example, we wait for the qubit to relaxed to its non-degenerate ground state, a Gaussian pulse with

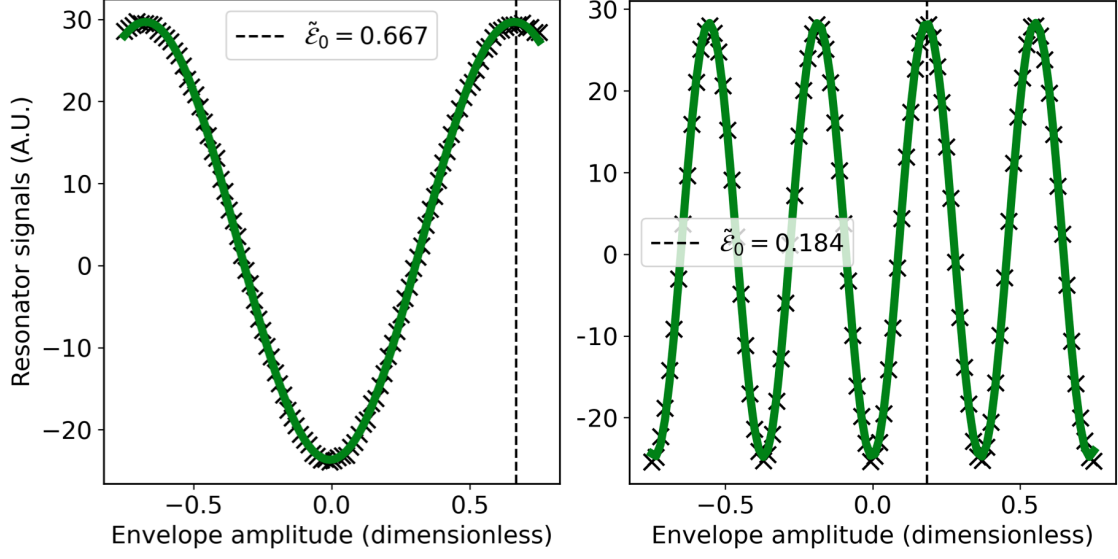


Figure 4.3: Coherent Rabi dynamics on (01) subspace. By driving a qutrit at frequency ω_{01} , we observed Rabi oscillation between the two eigenstates $|0\rangle$ and $|1\rangle$.

standard deviation $\tau/4$ results in the probability of measuring $|1\rangle$ to be

$$|p_e|^2 = \sin^2(\Omega_R \tau / 2), \quad (4.4)$$

hence the oscillatory behavior when drive field is turned on. As promptly stated, this corresponds to the rotation of the state vector around the x -axis on the Bloch sphere, a signature of X coupling between the spin-1/2 particle and the external field. As noted above, we care only about either the scaled duration if the amplitude is fixed; or the scaled amplitude if the duration is fixed. On IBM's hardware, it is customary to fix the duration and vary the amplitude. We consider two durations here, $\tau = 16$ ns and $\tau = 60$ ns. For each duration, we vary the amplitude from -0.75 to 0.75 (negative amplitudes correspond to counter-clockwise rotation on the Bloch sphere). The measured signal from the cavity is then again obtained using a heterodyne detection scheme. We determine $\tilde{\mathcal{E}}_0$ for subspace by fitting to the cosine function,

$$f(a) = A \cos(2\pi T \times a + B) + C, \quad (4.5)$$

where a denotes the amplitude, T is the period of the cosine, A, B, C are free parameters. Here, it is clear that the Rabi oscillations on (12) subspace are less fitted compared to that on (01). This is one of the current challenge present in our system. The fact that the cosine is not well-fitted implies that we would have coherent errors when using these pulses to realize actual single-qutrit gates. The shorter pulse duration $\tau = 16$ ns results in a well-behaved Rabi oscillation patterns for both subspaces, while the fitting variance for the $\tau = 120$ ns worsens significantly for the (12) subspace. For this reason, we pick the duration for both subspace to be $\tau = 16$ ns. The corresponding amplitudes for (01) and (12) are 0.667 and 0.352 respectively.

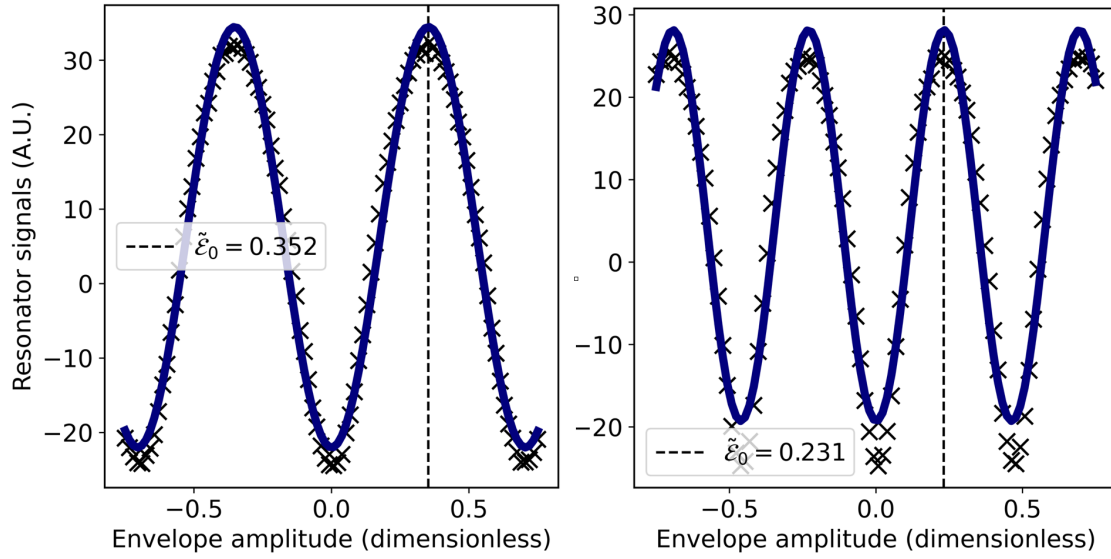


Figure 4.4: Coherent Rabi dynamics on (12) subspace. By driving a qutrit at frequency ω_{12} , we observed Rabi oscillation between the two eigenstates $|0\rangle$ and $|1\rangle$. The reason why the cosines are not well-fitted is postponed until Sec. 4.3.

4.3 Ramsey experiment

Previously, we determined the qutrit transition frequencies $\omega_{01}/2\pi$ and $\omega_{12}/2\pi$ spectroscopically. We now fine tuning these frequencies using a Ramsey experiment. The idea behind this experiment is to go into the frame where the drive frequency $\omega_d/2\pi$ is slightly detuned from the real frequency $\omega_{(s)}/2\pi$ by an amount δf . In this frame, the Bloch vector precesses around the quantization axis at the frequency δf . We first apply a $\pi/2$ pulse (a $\pi/2$ rotation around the x or y axis) to prepare an eigenstate of either σ_x or σ_y . Then if we wait for an amount δt , then the total precession around the equator is $2\pi f \delta t$. When the measurement is made, note that we are making the measurement in the frame rotating with the drive's frequency $\omega_d/2\pi$. Therefore, after waiting for δt and applying another $\pi/2$ pulse, we should receive a signal that is a function of $\delta t, \delta f$. In this case, the signal is the population of the higher energy state $|1\rangle$ or $|2\rangle$.

Once the signal is retrieved, it is a cosine function of waiting time δt . The oscillating pattern is caused by the detuning between the drive and the real transition frequency, δf . Here, we consider two different detuning value namely $\delta f_1 = 3$ MHz and $\delta f_2 = 10$ MHz. For (01) subspace, this δf_{01} is well resolved when we look at the spectral profile of the signal, see Fig. 4.5. The dominant frequency resembles a Dirac delta function peaks at $\delta f_{01} = 3.012$ MHz and $\delta f_{01} = 10.042$ MHz. Knowing that the tone sent was $\omega_d = 4.8505$ GHz (obtained from Section 4.1), we conclude that the fine-tuned transition frequency is less than 0.1 percent away from the frequency obtained from spectroscopy.

For the (12), the situation is much more complicated. We observe two peaks when looking at the spectral density, at 4.0161, 3.0121 and 11.0442, 10.0402 for $\delta f_{12} = 3$ MHz and $\delta f_{12} = 10$ MHz respectively. The 3.0121 MHz and 10.0402 MHz are the two expected peaks as in the case of (01), because we sent in a pulse that was 3 and 10 MHz detuned

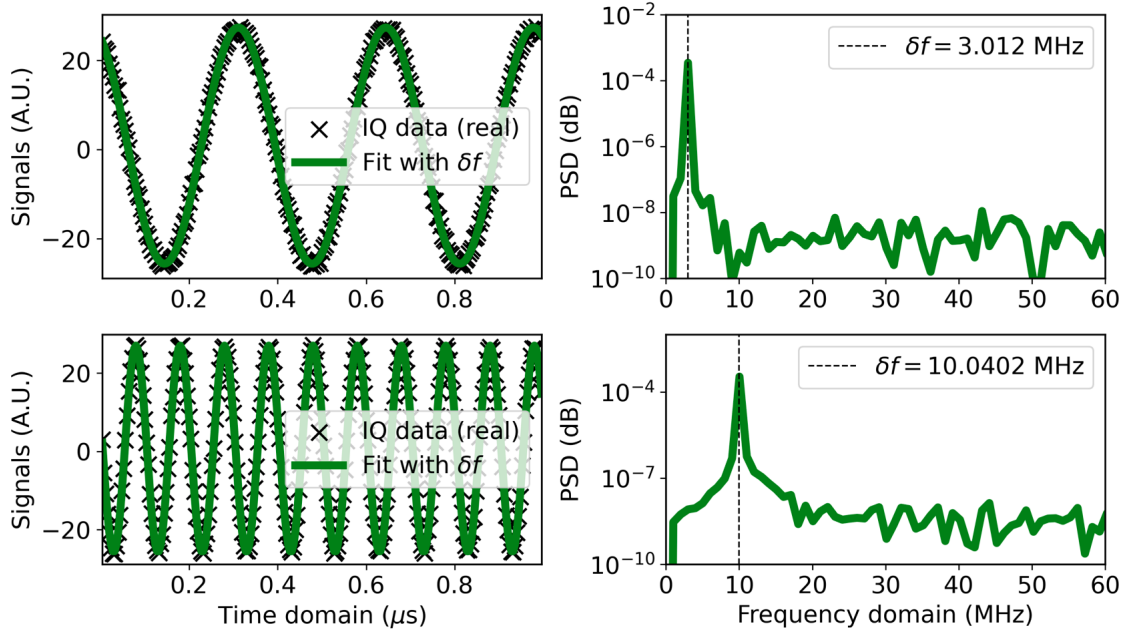


Figure 4.5: Ramsey oscillation on (01) subspace. (Top row, left) Ramsey signals in time domain using a pulse detuned 3 MHz away from the transition frequency obtained spectroscopically. The cosine is well approximated by a single-frequency signal. (Top row, right) Fast fourier transform of the signal. The peak is well-defined and looks like a Dirac delta function. (Bottom row, left) Ramsey signals in time domain using a pulse detuned 10 MHz away.

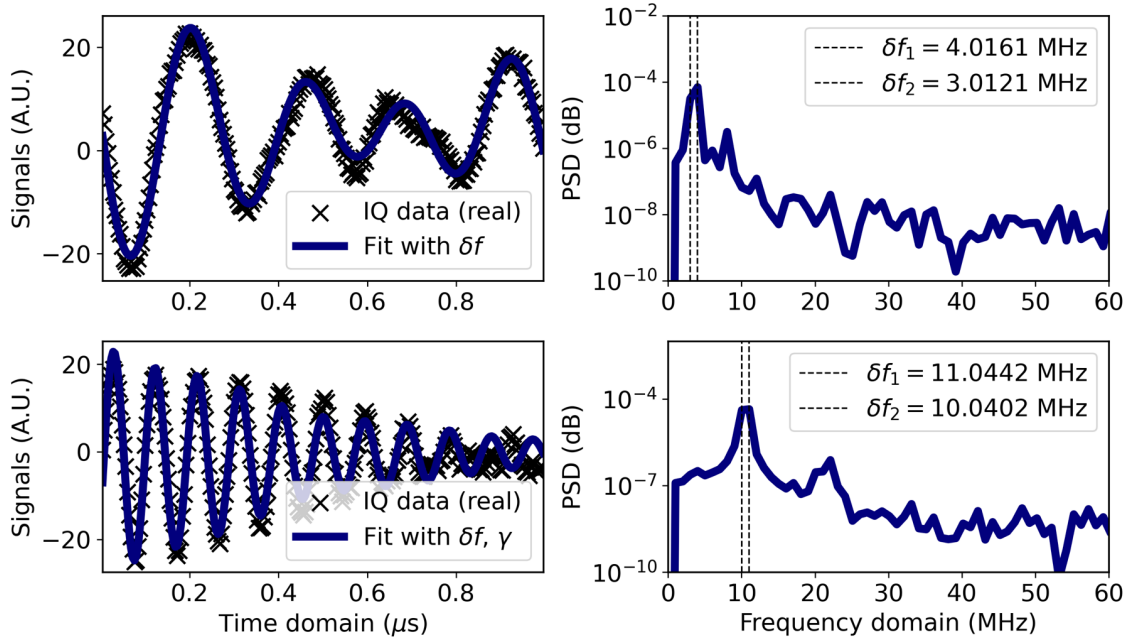


Figure 4.6: Ramsey oscillation on (12) subspace. (Top row, left) Ramsey signals in time domain using a pulse detuned 3 MHz away from the transition frequency obtained spectroscopically. The signal can be only fitted using the sum of two signals with a frequency difference of 1 MHz. (Top row, right) Fast fourier transform of the signal. The peak is split into two components at approximately 4 and 3 MHz. (Bottom row, left) Ramsey signal with pulse detuned 10 MHz away.

away from $\omega_d/2\pi = 4.5081$ GHz. The origin of these two peaks and how they affect gate fidelity are two active research directions within our group. Readers are advised to see [31, 33] for similar observations, where the gap was eventually attributed to quasiparticle tunneling events between the two superconducting islands.

Incoherent-limited control and benchmarking

To realize high-fidelity control over a single superconducting qutrit, one must overcome a plethora number of noise sources.

- Increased transversal-coupled decay due to bosonic enhancement. In qubit systems, the environmental noises coupled to the transversal direction are the main limiting factor of the T_1 time. In qutrit systems, this problem is more pronounced with an additional third energy level due to the effect of bosonic enhancement, which indicates that the relaxation rate is proportional to the number of photon (i.e. higher excitation states tend to decays faster with linear law to their photon number). The T_1 time of the (12) subspace is roughly half that of the (01) subspace.
- Dephasing due to charge noise on higher energy subspaces. This effect has been observed in Ramsey experiments in Sec. 4.3. While all transmon qubits available on IBM system are in the $E_J/E_C \gg 1$ regime that makes the charge dispersion ϵ_{01} on (01) subspace suppressed to zero, this ratio (typically 30 to 40) is not enough to suppress the charge dispersion on higher subspaces. In our case, the charge dispersion is estimated to be roughly 1 MHz, which in turn significantly limit coherent control over the (12) subspace. To this date, the peculiar structure of this noise remains a fundamental obstacle in realizing high-fidelity single qutrit gates. For this reason, we claim that the corrections we make in the following sections help us to push the gate fidelity as high as possible, limited by only the transmon qutrit's intrinsic incoherence.
- Phase advance in high-dimensional system. While the global phase in a qubit system can be safely ignored since it has no measurable effect on the computational basis states $|0\rangle, |1\rangle$, the situation for qutrit is much more complicated. In [29, 6, 22] the authors reported the so-called phase advance arising in the idling subspace when driving the other one. Since such phases disturb only the relative phase between qutrit's computational states, their effects are inelastic and do not destroy quantum phase coherence. The crucial observation is that advancing phases manifest themselves as accumulative relative phases of a *coherent superposition* of the idling

eigenstates ($\bar{s}$) while driving the target subspace (s). More importantly, previous works within our lab [29] have also point out that this phase advance is function of gate time t_g and anharmonicity α . However, the specific form of this function is not known and to correct for phase advance, one has to perform a specific type of experiment called random parameterized circuit. It is an exhaustive yet interesting problem that is being actively researched in our lab. Within the context of this thesis, the phase advance problem can be safely ignored because in the calibration procedures described here, to properly distinguish unitary phase errors encoded in qutrit population from advancing phases, we design gate sequences such that relative phases of any superposition formed are either destroyed through projective measurements or irrelevant to the measured population.

5.1 Errors in a superconducting qutrit

5.1.1 Phase errors

Virtual transitions outside of the computational subspace during driving, e.g. to the state $|3\rangle$ inflict an AC Stark shift on the qutrit's (12) subspace and result in a phase error. This phase error causes a quantum trajectory to deviate from the original path. For example, starting from the $|1\rangle$ state, a rotation $\mathcal{R}_X^{12}(\pi/2)$ does not bring us to the superposition $(|0\rangle + i|1\rangle)/\sqrt{2}$, but to a certain state $\mathcal{R}_X'^{12}(\pi/2)|1\rangle$. The unitary $\mathcal{R}'$ when expressed in circuit language is related to the original unitary as

$$\mathcal{R}_X'^{12}(\pi/2) = e^{i\phi} \mathcal{R}_Z^{12}(\epsilon_m) \mathcal{R}_X^{12}(\pi/2) \mathcal{R}_Z^{12}(\epsilon_m), \quad (5.1)$$

where ϕ is a general unknown phase factor that could potentially affect the logical state of the $|0\rangle$ state, $\mathcal{R}_{X/Z}$ are rotations realized from the Gell-Mann matrices 2.51.

The origin of the AC Stark shift can be assigned to the level repulsion between the $|1\rangle$ and $|2\rangle$ when driving the $|0\rangle$ and $|1\rangle$ subspace and vice versa. The level repulsion, in turn, causes a frequency shift on the transition frequency of the subspace. A qualitative explanation can be obtained by calculating the spectral density of a pulse using to realize our gates. Recall that in general a pulse can be described as a sum of two-quadrature (Eqn. 3.35,

$$\mathcal{E}_d(t) = \mathcal{E}_x(t) \cos(\omega_d t + \theta_0) + \mathcal{E}_y(t) \sin(\omega_d t + \theta_0), \quad (5.2)$$

$$= [\mathcal{E}_x(t) - i\mathcal{E}_y(t)] \exp(i\omega_d t), \quad (5.3)$$

where the phase has been absorbed into the amplitude. When we Fourier transform this envelope,

$$\mathcal{E}_d(\omega) = \int dt e^{-i\omega t} \mathcal{E}_d(t), \quad (5.4)$$

which showcases that even when $\omega_d = \omega_{01}$, we still observe frequency components near

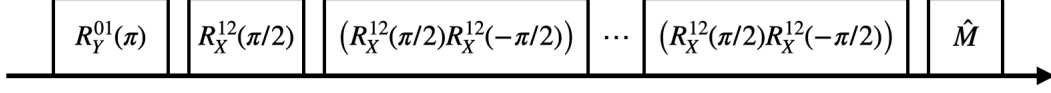


Figure 5.1: Pulse sequence used for measurement of phase error using APE protocol. The starting state is the ground state $|0\rangle$. The pair $[\mathcal{R}_X^S(\pi/2, \delta) \cdot \mathcal{R}_X^S(-\pi/2, \delta)]^n$ is repeated n times to modify the Ramsey fringe.

resonance to $\omega_{01} - \alpha = \omega_{12}$. When the transition-induced does not invert itself at the end of the gate time t_g , i.e. the population escapes the computational subspace, a leakage error occurs. If the transition-induced population transfer them back to the computational subspace, a phase error happens.

5.1.2 Measuring phase errors

To measure these errors, we extend the amplified phase error pulse (APE) sequence [24] to our qutrit system. The APE sequences in our testbed resemble a Ramsey fringe scheme in which n pseudo-identity operations are concatenated to accumulate phase errors ϵ_p on a prepared superposition of qutrit's eigenstates, followed by a final axis-varying $\pi/2$ pulse [See Fig. 5.1]. The sandwiched pairs of $\pi/2$ rotations,

$$[\mathcal{R}_X^S(\pi/2, \delta) \cdot \mathcal{R}_X^S(-\pi/2, \delta)]^n \approx \mathcal{R}_Z^S(2n\epsilon_p), \quad (5.5)$$

approximate an effective Z gate where the phase errors δ are amplified with increasing n . Therefore, the phase error ϵ_p is amplified by $2n$ for n successive application of the pair $(\mathcal{R}(\pi/2)\mathcal{R}(-\pi/2))^n$. We can directly probe this phase by applying an additional axis-varying $\pi/2$ pulse at the end of the circuit.

5.1.3 Correcting phase errors using DRAG

The so-called DRAG procedure (Derivative Removal of Adiabaticity Gate) seeks to combat phase errors by scaling the quadrature component $\mathcal{E}_y(t)$. The quadrature component is defined to be the time derivative of the in-phase component, so that the DRAG waveform is

$$\mathcal{E}_{\text{DRAG}}(t) = [\mathcal{E}_x(t) - i\beta\dot{\mathcal{E}}_x(t)] e^{i\omega_d t}, \quad (5.6)$$

where β is the scaling parameter. Intuitively, the Fourier transform is for a DRAG envelope is

$$\mathcal{E}_{\text{DRAG}}(\omega) = \int dt e^{-i\omega t} [\mathcal{E}_x(t) - i\beta\dot{\mathcal{E}}_x(t)] e^{i\omega_d t}, \quad (5.7)$$

$$= [1 - \beta(\omega - \omega_0)] \int dt e^{-(\omega - \omega_d)t} \mathcal{E}_x(t). \quad (5.8)$$

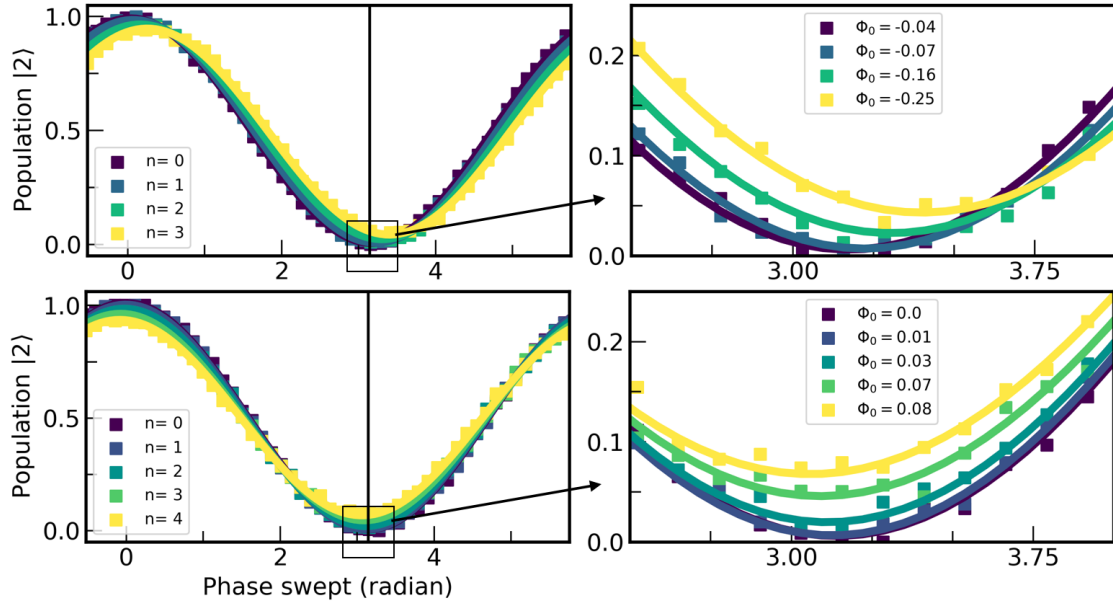


Figure 5.2: Ramsey fringes obtained from the APE sequence. (Top row, left) APE sequence before DRAG correction. The population of $|2\rangle$ is a function of the rotation axis of the final $\pi/2$ pulse. For each repetition $n = \{0, 1, 2, 3\}$ the cosines (fringes) appears to further de-synchronized from each other. (Top row, right) Zoomed into the point at $\phi = \pi$. Fitted initial phases reveal a maximum of 0.21 radians of phase difference. (Bottom row, left) APE sequence after adding DRAG correction. (Bottom row, right) Fitted initial phases reveal a maximum of 0.08 radians of phase difference. A 60 percents quantified reduction.

The original theory of AC Stark shift tells us that the frequency shift is proportional to $-\mathcal{E}_x^2(t)/2\alpha$, where α is the anharmonicity. It is also clear from Eqn. 5.7 that the derivative term shifts the qubit's frequency by $-2\beta\mathcal{E}_x^2(t)$. Hence, the total frequency shift is $-(1 + 2\beta)\mathcal{E}_x^2(t)$. In theory if we choose $\beta = -0.5$, then the shift vanishes. A more rigorous treatment of DRAG theory is presented in [28, 15].

Here, due to pulse distortion and parameter shift, $\beta = -0.5$ is not the optimal choice. We therefore rely on the amplified phase error pulse sequence (APE) above to find the optimal scaling parameter β . Note the behavior of a phase-error-free sequence in Fig. 5.2 (top row). It follows that the cosine function (with respect to phase swept) should be in phase with each other in error-free case. Thus, we vary β such that it minimizes the difference in initial phases of the cosine functions. Before correction, $\max|\Phi_k - \Phi_l| = 0.21$; using the optimal DRAG value β , this number reduces to $\max|\Phi_k - \Phi_l| = 0.08$, quantifying a nearly 60% improvement in phase error. Due to the nature of Hamiltonian parameter drifts, the plots for obtaining optimal scaling parameter β is not shown here.

5.1.4 Over/under-rotation errors

Mistorated errors result from the driving strength $\tilde{\mathcal{E}}_0$ not being properly calibrated. This could either come from the finite accuracy of a Rabi oscillation experiment or nonlinearity-induced pulse distortion along the drive transmission line (See [23] for a comprehensive study). A signature of this type of error is that they add coherently with each time

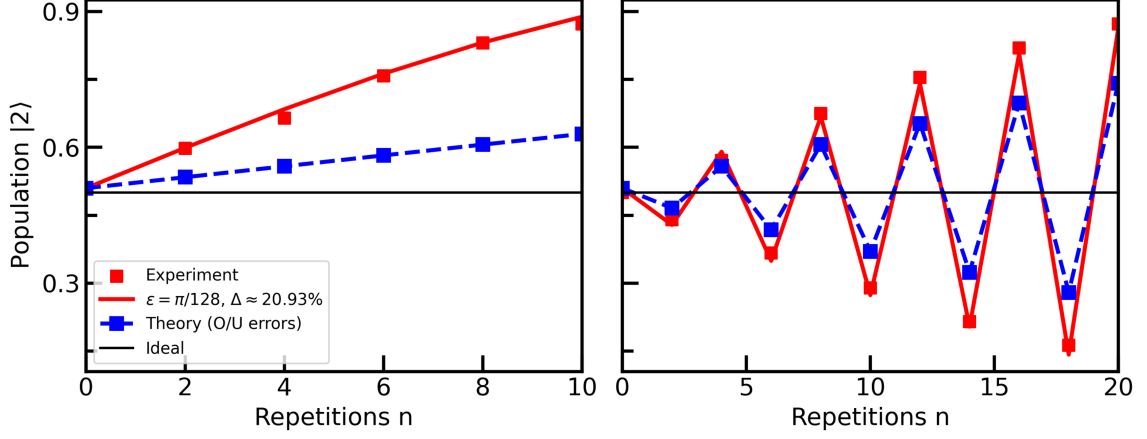


Figure 5.3: Calibration of over/under-rotations. (Left) Oscillating pattern of $\langle \sigma^{12} \rangle = P(|2\rangle)$ for the gate $\mathcal{R}_X^{12}(\pi)$. (Right) Oscillating pattern of $\langle \sigma^{12} \rangle = P(|2\rangle)$ for the gate $\mathcal{R}_X^{12}(\pi/2)$. The fitted value of over rotation, approximately $\pi/128$ is feed-forward directly to further experiments to correct for misrotated errors.

the gate is applied, say n times. Then, assumed a general form of the unitary error, $\exp(-i\frac{\epsilon_m}{2}\hat{r} \cdot \vec{\lambda})$, where $\epsilon_m, \hat{r}, \vec{\lambda}$ are the error angle, axis of error and the corresponding Gell-Mann element. If the misrotated angle is a small parameter, $\epsilon_m \ll 1$, then a sequence of n gates can be expanded to second order in ϵ_m as

$$U^n = I - i(n\frac{\epsilon_m}{2})\hat{r} \cdot \vec{\lambda} - (n(2n-1))\frac{\epsilon_m^2}{4}(\hat{r} \cdot \vec{\lambda})^2 + O(\epsilon_m^3). \quad (5.9)$$

The error can be further characterized by introducing the concept of averaged gate fidelity through all input states in the qutrit's Hilbert space,

$$F_g = \int d\psi \operatorname{tr} [U_{\text{ideal}}^n |\psi\rangle\langle\psi| (U^\dagger)_{\text{ideal}}^n U_{\text{actual}}^n |\psi\rangle\langle\psi| (U^\dagger)_{\text{actual}}^n], \quad (5.10)$$

then the measured fidelity can be expressed in terms of n and ϵ ,

$$\alpha = 1 - \frac{n(2n-1)\epsilon_m^2}{3}, \quad (5.11)$$

which depends quadratically on n . This result is also observed in [39] for qubit's case. In practice, F_g can be measured using the theory of randomized benchmarking. But conducting a randomized benchmarking experiment could be costly. Here, we seek a simpler way of measuring ϵ . Under the assumption that the error is purely misrotated, then the expectation value of the two observables $\sigma_{01} = |0\rangle\langle 0| - |1\rangle\langle 1|$ and $\sigma_{12} = |1\rangle\langle 1| - |2\rangle\langle 2|$ exhibits oscillating pattern (zig-zag-like) between -1 and $+1$. If the gate is $\mathcal{R}_X^S(\pi/2)$ where S admit either (01) or (12), then the expectation values can be fitted directly into

$$\langle \sigma_S \rangle = a + \left(\frac{1}{2}(-1)^n \cos \left(\frac{\pi}{2} + 2n\epsilon_m \right) \right), \quad (5.12)$$

where ϵ_m is the error parameter of interested and a is a free parameter. The experiment result is obtained with Gaussian pulse, without DRAG correction (See Fig. 5.3). The error parameter ϵ_m is estimated to be roughly $\pi/128$ radians.

5.2 Randomized benchmarking

Having established how to push the limit of single qutrit control on a IBM system, we will now benchmark the performance of our single qutrit gates. We first review the technique of Clifford-based randomized benchmarking, then generalize it to our qutrit to estimate the total gate fidelity.

5.2.1 Qubit randomized benchmarking

So far we have assume that the two dominant sources of error in our qutrit system are the misrotated errors and phase errors. It is however lacking of a single, simple figure of merit to determine the actual improved performance after making DRAG and over/under rotation corrections. One way to characterize this is to perform quantum process tomography. However, it is quantum process tomography (QPT) is often costly, and the gates used in QPT themselves are erroneous gates. To overcome these drawbacks, we turn to the theory of randomized benchmarking using random unitaries to obtain averaged fidelity. The core principle behind this idea was that when an error process $\mathcal{E}(\rho)$ is averaged over the uniform space of unitaries, called the *Haar measure*, the result is a depolarizing channel $\mathcal{E}_\lambda$, which maps any pure state ρ to the maximally mixed state I .

$$\mathcal{E}_\lambda(\rho) = \sqrt{\frac{\lambda}{4}} X^\dagger \rho X + \sqrt{\frac{\lambda}{4}} Y^\dagger \rho Y + \sqrt{\frac{\lambda}{4}} Z^\dagger \rho Z + \sqrt{\frac{1-3\lambda}{4}} I, \quad (5.13)$$

where X, Y, Z are the Pauli matrices in the qubit case; the process is fully parameterized by a single number λ , which determines the contraction rate of a pure state ρ towards a mixed state $\mathcal{E}_\lambda(\rho)$. We note that in a depolarizing channel, the error rate λ is identical for all Pauli errors. The original paper [?] proposed choosing a random unitary U and construct a motion reversal sequence, namely $UU^\dagger$. Let us start from the ground state $|0\rangle$, then the probability measuring the system in its ground state after the motion reversal sequence is a function of the strength of the error, $p(|0\rangle) = |\langle 0 | \tilde{U} \tilde{U}^\dagger | 0 \rangle|^2$, where $\tilde{U}$ is the faulty operator that we wish to characterize. Doing this experiment just one time is not realistic, because λ is intrinsically a probability that an error will happen. We therefore need to repeat the experiment many times to obtained the statistics. The full protocol to perform a randomized benchmarking experiment is as follow,

1. Draw a set of m random unitaries, $U_1, U_2, \dots, U_m$. This set determines the circuit depth m .
2. Construct their motion reversal operations, $U_1 U_1^\dagger, U_2 U_2^\dagger, \dots, U_m U_m^\dagger$.
3. Prepare a ground state $|0\rangle$.

4. Apply the sequence $\prod_{j=1}^m U_j U_j^\dagger$ to the ground state.
5. Measure the ground state population, $p(|0\rangle) = |\langle 0 | \left(\prod_{j=1}^m U_j U_j^\dagger \right) |0\rangle|^2$.
6. Repeat steps 1 – 4 many times to get the statistics. For each repetition choose different circuit depth m .

As the circuit depth m increases, the averaged effect of the sequence tends closer to that of a depolarizing channel. If one were to plot the population $p(|0\rangle)$ with respect to circuit depth m , the plot is then an exponential decay. This exponential decay can be fitted such that we absorb the information about preparation and measurement errors into the free parameter, leaving only the information about the noisy process in the decay constant.

The most challenging part of the original protocol is that we need to draw unitaries randomly from the Haar measure on the $SU(2)$ group. This is in fact an infinite number of unitary operators. The large cardinality properties of this set makes it impractical to implement on hardware. To overcome this, we can draw only representative elements from the Haar measure. It turns out that there exists such a representative set of $SU(2)$ called *Clifford group*, denoted by $\mathcal{C}$, which is a unitary-3 design¹. To better appreciate the Clifford group, we circle back to the averaged fidelity defined in Eqn. 5.10. Here, we are integrating over the set of all possible states $|\psi\rangle$. We modify this expression to integrate over the set of all possible unitaries U , suppose that we start from the ground state $|0\rangle$. For the protocol described above, a ground state $|0\rangle$ is then mapped to a random state using the randomly drawn unitary U ; in such case, the averaged fidelity is

$$F_g = \int d\mu(U) \operatorname{tr} \left[\mathcal{E}(|0\rangle\langle 0|) V^\dagger U^\dagger |0\rangle\langle 0| V U \right], \quad (5.14)$$

where V is the ideal unitary and $\mathcal{E}$ is the actual unitary realized in the experiment. This is called twirling over the channel $\mathcal{E}$. This integral is exceedingly hard to compute, if not impossible. Using a unitary-2 design, we can however equate it to

$$F_g = \frac{1}{K} \sum_{j=1}^K \operatorname{tr} \left[\mathcal{E}(|0\rangle\langle 0|) V^\dagger U_j^\dagger |0\rangle\langle 0| U_j V \right]. \quad (5.15)$$

We stress that this is not an approximation. The two F_g are demonstrably equal to each other.

For completeness, I write down here the modified protocol to perform a randomized benchmarking using the Clifford group.

1. Prepare a ground state $|0\rangle$.
2. Apply a sequence of m Clifford operators, $\prod_{j=1}^m C_j$ to the ground state.
3. Apply a final gate to the sequence. The final gate is defined such that the whole ideal sequence returns the identity operator, $C_{m+1} \prod_{j=1}^m C_j = I$.

¹Note that a unitary- d design is by definition a unitary- t design, for all $0 < t \leq d$.

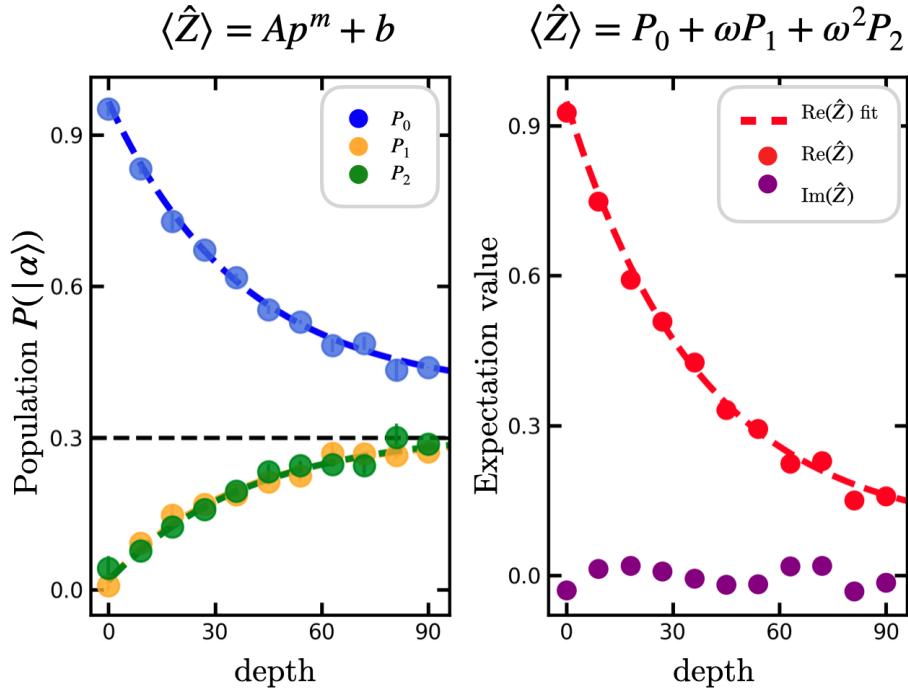


Figure 5.4: Qutrit randomized benchmarking. (Left) The signature decay of a qutrit randomized benchmarking experiment when twirling through the Clifford group. For a circuit depth of 90, the populations of all eigenstates tend to $1/3$, indicating a maximally polarized state $I/3$. (Right) The expectation value of the non-Hermitian operator Z defined in Eqn. 5.18. The fitted decay parameter p indicates an averaged Clifford gate fidelity of $F = 0.989$.

4. Measuring the ground state population $p(|0\rangle)$.
5. Repeat the steps from 1 to 4 multiple times, with each repetition choosing a new m circuit depth.

This protocol is similar to the motion reversal version, except that the inverse element is calculated all the way in the end; furthermore, this element can be, by definition, efficiently found because the Clifford group is closed under multiplication.

5.2.2 Qutrit randomized benchmarking

The idea of randomized benchmarking can be extended to the qutrit case. We need to find the "representative set" for $SU(3)$, which is the single-qutrit Clifford group C_3 . This group is generated by two generators, namely Hadamard gate H and phase gate S ,

$$H = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}, \quad (5.16)$$

$$S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \omega \end{pmatrix}, \quad (5.17)$$

where ω is the primitive d root of unity. The group order is 216. We note that using the elementary gate set defined in Eqn. 2.51, we can in principle generate the Hadamard gate (with at most two physical gates $\mathcal{R}_X^S(\pi/2)$) and the phase gate, which is diagonal (and therefore virtual, See [26]). Using the decomposition given in Eqn. 2.59, the Clifford group elements are therefore broken down to a set of instruction whose main ingredient is the two elementary gates $\mathcal{R}_X$ and $\mathcal{R}_Z$. To extract the figure of merit, in this case, the decay constant, we fit the decay to a single exponential [27],

$$\langle Z \rangle(m) = ap^m + b, \quad (5.18)$$

where m is the circuit depth, a, b are free parameters that absorb in the state preparation and measurement errors. The observable Z is defined as

$$\langle Z \rangle = P(|0\rangle) + \omega P(|1\rangle) + \omega^2 P(|2\rangle), \quad (5.19)$$

where P is the population of the computational basis states $|0\rangle, |1\rangle, |2\rangle$. The results of qutrit randomized benchmarking for our calibrated gates using DRAG corrections are presented in Fig. 5.4. The fitted decay p indicates an averaged Clifford gate fidelity of $F = 0.989$.

Interleaved randomized benchmarking

We note that the averaged fidelity obtained in Eqn. 5.15 is the *averaged* over many m and *averaged* over the whole Clifford group. This figure of merit is sometimes not desired if we have a specific gate that need to be characterized, instead of *the whole* group. This problem is addressed by interleaved randomized benchmarking, where one interleaves the gate of interest between the Clifford twirls. Then, comparing with the case where one does not interleave the gate reveals the gate error. Let p_i and p are the decay rate with and without interleaving, then the gate error is estimated to be

$$r_g = \frac{2}{3} \left(1 - \frac{p_i}{p} \right). \quad (5.20)$$

At the time of writing this thesis, interleaved randomized benchmarking experiments are being carried out in the lab.

CHAPTER 6

Conclusion

In this thesis, we have explored the control of a three-level system, or from a quantum information perspective a qutrit realized by engineering a superconducting transmon circuit to operate in within the computational subspace spanned by its first three eigenstates. Theory-wise, starting from the Hamiltonian of the qutrit, we established the universal control theory on a single superconducting qutrit system by showing that any unitary in $SU(3)$ group can be decomposed into a sequential moves in each individual subspaces $\{|0\rangle, |1\rangle\}$ and $\{|1\rangle, |2\rangle\}$, which can be further decomposed into just two elementary rotations, namely rotations about the x and z axes. Experiment-wise, we have characterized the coherence of the individual subspaces and found that the incoherence source comes mostly from the $\{|1\rangle, |2\rangle\}$ subspace, indicating by the splitting of the ω_{12} transition frequency measured in a Ramsey experiment. We have also identified the major source of unitary errors being the phase errors, resulted from the AC Stark shift when addressing an individual subspace. The phase errors are then amplified by refined use of the so-called amplified phase error (APE) sequence. Derivative Removal of Adiabaticity Gate (DRAG) corrections were used to combat this phase error, which results in a 60 percent reduction compared to non-corrected case. Finally, we have also performed Clifford-based qutrit randomized benchmarking, resulting in an averaged gate fidelity of 98.8 percent.

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Contents

A Proof of Mathematical Theorems	64
A.1 Parity of a permutation	64

Proof of Mathematical Theorems

This appendix contains the proofs of all claims made in main body of the dissertation.

A.1 Parity of a permutation

In its simplest form, permutation refers to the action of rearranging elements in a fixed set.

Definition 1 (Permutation). *A permutation of length n is an invertible function from a set to the same set*

$$\{1, 2, \dots, n\} \longrightarrow \{1, 2, \dots, n\} \quad (\text{A.1})$$

Note that a function is invertible if and only if that function is simultaneously surjective and injective, i.e. it's a bijective map. This ensures that a permutation does not create nor destroy previous elements. Let us look at an example,

$$\{1, 2, 3, 4, 5, 6\} \longrightarrow \{4, 6, 1, 2, 5, 3\} \quad (\text{A.2})$$

There sure exists a bijective map between the two sets. We call this a permutation. Such permutation can be describe by a $2 \times n$ matrix, also known as Cauchy's two-line notation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 6 & 1 & 2 & 5 & 3 \end{pmatrix} \quad (\text{A.3})$$

or can be referred to as $\sigma(1) = 4, \sigma(2) = 6, \sigma(3) = 1, \sigma(4) = 2, \sigma(5) = 5, \sigma(6) = 3$. Of course, for a set of length n , the number of possible permutations is $n!$. The set of all permutations of length n is denoted as S_n . We now discuss the definition of an *inversion*.

Definition 2 (Inversion). *Given a permutation σ of length n , a pair $[i, j]$ where $1 \leq i < j \leq n$ is called an inversion if $i < j$ and $\sigma(i) > \sigma(j)$.*

Let us look at a minimal example. Let us consider the previous permutation σ such

that

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 6 & 1 & 2 & 5 & 3 \end{pmatrix} \quad (\text{A.4})$$

How many inversions we have? The first one is $[1, 3]$, since $\sigma(1) = 4 > 1 = \sigma(3)$ while $1 < 3$. Similarly, $[1, 4]$, $[1, 6]$, $[2, 3]$, $[2, 4]$, $[2, 5]$, $[2, 6]$, $[5, 6]$ are the other inversions. We have 8 inversions in total. Let us call the number of inversions in a permutation μ . Let us now define the parity.

Definition 3 (Parity). *The parity of a permutation σ is $(-1)^\mu$. We call σ even (or odd) if the μ is even (or odd).*

We will further establish some important properties of permutations using the concept of *transposition*. A transposition is a *cycle* of length 2, meaning that it swaps two element (a, b) . Here's an example in which we have a set of 6 elements,

$$(4, 6) \in S_6 \quad (\text{A.5})$$

meaning that we're swapping 4 and 6. This transposition can be effectively written as, using the notation of Cauchy's two-line notation

$$(4, 6) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 6 & 5 & 4 \end{pmatrix} \quad (\text{A.6})$$

where we're swapping non-adjacent transposition, 4 and 6. Intuitively, we can construct non-adjacent from adjacent transposition $(i, i + 1)$. More generally,

Theorem 2 (Adjacent transposition is universal). *Every permutation $\sigma \in S_n$ could be factored into a group of product of adjacent transpositions.*

We take this theorem for granted. What does this theorem mean?

Remark 6. *If σ has μ inversions, then σ could be factored as a product of μ adjacent transpositions.*

As a straightforward example, let us take the σ permutation above.

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 6 & 1 & 2 & 5 & 3 \end{pmatrix} \quad (\text{A.7})$$

Let us factor the permutation into a series of adjacent transpositions. One algorithmic way of doing this is to look for inversions. First, we have $(2, 3)$, hence the permutation can be factored into

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 1 & 6 & 2 & 5 & 3 \end{pmatrix} (2, 3) \equiv \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 1 & 6 & 2 & 5 & 3 \end{pmatrix} t_{2,3} \quad (\text{A.8})$$

Next, we have $(1, 2)$, hence the permutation is further factored into

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 4 & 6 & 2 & 5 & 3 \end{pmatrix} t_{1,2} t_{2,3} \quad (\text{A.9})$$

Repeating this process several times, we obtained the permutation σ in terms of adjacent transpositions only

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 4 & 2 & 6 & 5 & 3 \end{pmatrix} t_{3,4} t_{1,2} t_{2,3} \quad (\text{A.10})$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 4 & 2 & 5 & 6 & 3 \end{pmatrix} t_{4,5} t_{3,4} t_{1,2} t_{2,3} \quad (\text{A.11})$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 4 & 2 & 5 & 3 & 6 \end{pmatrix} t_{5,6} t_{4,5} t_{3,4} t_{1,2} t_{2,3} \quad (\text{A.12})$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 4 & 2 & 3 & 5 & 6 \end{pmatrix} t_{4,5} t_{5,6} t_{4,5} t_{3,4} t_{1,2} t_{2,3} \quad (\text{A.13})$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 4 & 3 & 5 & 6 \end{pmatrix} t_{2,3} t_{4,5} t_{5,6} t_{4,5} t_{3,4} t_{1,2} t_{2,3} \quad (\text{A.14})$$

$$= t_{3,4} t_{2,3} t_{4,5} t_{5,6} t_{4,5} t_{3,4} t_{1,2} t_{2,3} \quad (\text{A.15})$$

where e being the identity operation (it does nothing to the set). Although this process is not unique, i.e. we can reconstruct σ through another even number of transpositions, we will never be able to construct σ through an odd number of transpositions. This is in agreement with the definition of the parity of a permutation.

Now, it's clear that through a process of $k = 8$ adjacent transpositions we can obtain σ , which has $\mu = 8$ inversions, so there's clearly a relationship between the parity and the number of transpositions, and it's the same as the relationship between the parity and μ (kind of obvious),

$$(-1)^\mu = (-1)^k \quad (\text{A.16})$$

This implies that the parity is preserved through the process of decomposition into adjacent transpositions, as observed above. At this point, we know what the parity of a permutation is. One ineffective way of finding the parity is to count for the number of inversions μ in the given parity σ , then the parity $(-1)^\mu_\sigma$ is even (or odd) if and only if μ is even (or odd, respectively).